

Generalized-Hukuhara Subgradient-Based Neurodynamic Approach to Nonsmooth Constrained Distributed Interval-Valued Optimization

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Abstract—In this article, we propose a projection-type neurodynamic approach for solving a class of nonsmooth constrained distributed optimization problems with interval-valued local objective functions over a network of vertices. We introduce the concept of generalized-Hukuhara subgradient (gH -subgradient) and differential inclusion system in the algorithm framework, aiming to cope with the confusion that objective functions may not be gH -differentiable, and further deriving the efficiency condition of the solution of the problem by employing the geometry relationship between the gH -subgradient and the normal cone. The boundedness and convergence of the state solution of the proposed neurodynamic algorithm are provable, and the conclusion that the state outputs of all vertices reach consensus on the efficient solution of the problem can be guaranteed by Lyapunov function method and Byrnes–Martin integral invariant principle. The designed neurodynamic model is simple in

structure, avoids the calculation of matrix inversion and the introduction of Lagrange multipliers, and is suitable for solving general distributed optimization problems with interval-valued functions and constraints. Finally, the simulation applications of network utility, power flow balancing, and gas transportation show the viability of the proposed algorithm.

Impact Statement—This work focuses on addressing a class of interval-valued optimization problems in a distributed manner in uncertain environments, where the local objective function being considered may not be gH -differentiability. From the perspective that neural network models can handle complex and uncertain optimization problems, we propose a projection neurodynamic algorithm to solve such distributed optimization problems. This algorithm incorporates the concepts of gH -subgradient and differential inclusion, overcomes the trouble of the local objective function not being gH -differentiable, and the convergence performance can be rigorously guaranteed. This work achievement provides a continuous and universal algorithm framework for solving constrained interval-valued optimization problems, and extending the application of neural networks and their dynamic potential in solving uncertain and complex optimization problems.

Index Terms—Differential inclusion, distributed optimization, generalized-Hukuhara subgradient, interval-valued functions, neurodynamic approach.

I. INTRODUCTION

WITH the rapid development of computer science, communication, automation, and other fields, as well as the continuous innovation of sensing technology, the advantages of solving complex network optimization problems in a distributed way are more and more prominent [1]. Distributed optimization is of great value for resource allocation [2], [3], economic scheduling [4], energy management [5], [6], etc. The solution of such problems usually depends on algorithm design, task division, communication protocol, vertex coordination, and other important conditions [7].

Reasonable distributed optimization algorithms can make full use of computing resources in distributed environment, speed up the solving process, and have good performance in dealing with large-scale and complex optimization problems [8], [9], [10], [11], [12], [13], [14]. For instance, Duchi et al. [8] proposed a distributed algorithm by means of the average of dual subgradients for a class of discrete optimizations with only local communication and derived the corresponding bounds

Received 9 October 2025; revised 15 December 2025; accepted 9 March 2026. Date of publication 18 March 2026; date of current version 31 August 2026. This work was supported in part by the National Natural Science Foundation of China under Grant 62202071, Grant 62302068, and Grant 62576053; in part by the Guangxi Science and Technology Program under Grant AD23023001; in part by the Natural Science Foundation of Chongqing under Grant CSTB2023NSCQ-LZX0149 and Grant CSTB2024NSCQ-MSX1087; in part by the Science and Technology Research Program of Chongqing Municipal Education Commission under Grant KJQN202501553; and in part by the Fundamental Research Funds for the Central Universities under Grant 2023CDJKYJH019. This article was recommended for publication by Associate Editor Tingwen Huang upon evaluation of the reviewers' comments. (Corresponding author: Xiaofeng Liao.)

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Digital Object Identifier 10.1109/TAI.2026.3675069

related to the convergence rates. Nedić et al. [9] came up with an algorithm distributed, which combined the inexact gradient operator and gradient tracking. Aforementioned algorithms are used for solving unconstrained optimization problems, while in general, the optimization problems encountered are constrained. Zhu et al. [10] proposed a projected primal-dual dynamic method to deal with distributed convex optimization with general constraints and proved that the algorithm asymptotically converges to the optimal point. Unlike the former, Wang et al. [11] introduced a distributed projected dynamic system with output feedback for solving a class of nonsmooth coupled-constrained optimization problems. Liu et al. [12] developed a distributed projection algorithm involving subgradient to solve constrained lasso regression problem by using a proper step size. Some distributed algorithms are also implemented over directed graphs. In particular, Gharesifard and Cortés [13] devised a distributed dynamic approach to solve convex optimization problems over strongly connected weight-balanced graphs by introducing suitable parameters, and for the case where time-varying and possibly random directed graphs, Saadatniai et al. [14] designed a novel distributed algorithm for minimizing a sum of strongly convex functions.

In general, the previously considered distributed optimization problems are based on the assumption that the local objective function is a deterministic value, and in the specific production practice, the objective function of the optimization problem may be an interval value due to a variety of uncertainties [15], [16]. Recently, some theoretical results on interval-valued optimization have been presented. Using the concept of gH -derivative, Osuna-Gómez et al. [17] derived an optimality condition for the optimization of interval-valued functions from stationary or critical points. Villanueva et al. [18] established a sufficient and necessary condition for the optimality of the optimization problems, which included interval-valued objective functions and real-valued inequality functions. For the interval-valued variational control, Treanță [19] discussed the relationship between the LU -optimal solution and the saddle-point of the corresponding Lagrange functional. Ghosh et al. [20] presented the concepts of gH -directional derivative, gH -Gâteaux derivative, and gH -Fréchet derivative about interval-valued functions.

While the research on optimal theory has made progress, the design and development of interval-valued optimization algorithm is still relatively insufficient. Arjmandzadeh et al. [21] equivalently transformed a random linear optimization problem with intervals into a convex cone programming problem and proposed a neurodynamic method with Lyapunov stability. Li et al. [22] presented a single-layer neural network approach to solve a class of linear constrained interval-valued optimization problems. Liu et al. [23] proposed a neurodynamic algorithm for solving a nonconvex interval-valued optimization problem constrained by partial order and linear equality. To the best of our knowledge, these are existing continuous-time algorithms for solving interval-valued optimization problems.

As a mainstream continuous-time optimization tool, neurodynamic methods shaped by recurrent neural networks have good performance in solving complex optimization problems,

and some of them are used to solve variational inequalities, mixed-integer programming, pseudomonotone projection, and absolute value equations based on the idea of gradient descent [24], [25], [26], [27]. With the development of nonsmooth analysis, the concept of subgradient and subdifferential has been widely used in the modeling of neurodynamics [28], [29], [30], [31], [32], [33], [34], [35], [36], [37], [38]. Li and Yan [29] used the method of subgradient-based neurodynamic to solve the l_1 -minimization problems in sparse signal reconstruction. Liu and Qin [30] developed a neurodynamic approach involving auxiliary function to optimization problems constrained by affine equality and inequality. For the solution of complex distributed optimization problems, the neurodynamic methods also show strong practicality. With the theory of neurodynamic systems, He et al. [31] designed a distributed approach for time-varying function optimization problems subject to inequality constraints. Liu et al. [32] presented a collective neurodynamic model to solve general distributed convex optimization. In view of the nonsmooth convex and strongly convex optimization problems, Zhao et al. [34] presented two neurodynamic approaches that are smoothed by projection operators. In [36] and [37], the proposed neurodynamic approaches are used for the distributed optimization problems with inequality constraints and general constraints, respectively. Xia et al. [38] created a collaborative neurodynamic system to address a class of distributed optimization problems with nonconvex functions.

On the basis of the observation of the above studies, it can be found that the neurodynamic approaches have good results in solving distributed optimization problems, and they have also been preliminarily applied to the solution of several kinds of interval-valued optimization problems. Considering the uncertainty of research problems in real scenarios, the study of distributed optimization with interval-valued local objective functions is of great value. Therefore, we present a neurodynamic approach based on gH -subgradient to solve the distributed interval-valued convex optimization problem in undirected connected networks. The main contributions of this work are as follows.

- 1) This article focuses on solving a class of constrained distributed convex optimization problems with interval-valued local objective functions, which have not been studied before. Compared with the research of real-valued distributed optimization problems, we do not assume the strong convexity of local objective functions [12], [14], [34], and do not restrict the differentiability of inequality constraint functions [31]. Therefore, the proposed neurodynamic approach can solve a broader range of distributed optimization problems.
- 2) In the past, the establishment of the optimal theory of interval-valued optimization was also based on the assumption that the objective function is gH -differentiable and the inequality constraint functions are differentiable [17], [18], while our work generalizes them to the nonsmooth case. Furthermore, the introduction of gH -subgradient reflects the innovation of dynamical methods.
- 3) In the design of the algorithm, we avoid the procedure of finding inverse matrices in the previous models [30],

[32], so that the calculation amount of the algorithm is reduced accordingly. Meanwhile, compared with the existing models involving Lagrange multipliers [10], [35], [36], our algorithm does not introduce auxiliary variables to deal with equality or inequality constraints.

- 4) The convergence of the proposed gH -subgradient-based neurodynamic algorithm has undergone rigorous mathematical proof and can be guaranteed on real vector space containing the family of all closed intervals. We verify the availability and feasibility of the algorithm on network utility maximization, power flow balancing, and natural gas pipeline transportation problems with interval-valued local costs.

Structure: The rest of this article is structured as follows. Section II presents some basic concepts. Section III gives the formulation of constrained distributed interval-valued optimization problem. In Section IV, a neurodynamic approach is proposed, and the efficiency condition is established. Section V analyzes the convergence and output consensus of the algorithm. In Section VI, several applications verify the relevant theoretical results. The conclusions and future prospects are contained in Section VII.

Notation: Define $\mathbb{R}$ is the set of real numbers, then $\mathbb{R}^n$ and $\mathbb{R}^{m \times n}$ are represent n -dimensional real Euclidean space and $m \times n$ -dimensional real vector space of matrices, respectively. For any $x = [x_1, x_2, \dots, x_n]^T \in \mathbb{R}^n$, denote $\|x\|$ and $\|x\|_1$ as Euclidean norm and l_1 -norm, respectively. Additionally, we use $[z_i]_{i=1}^N$ to denote the column vector formed in the order of vectors (or elements) $z_1, z_2, \dots, z_N$. For matrices $A \in \mathbb{R}^{N \times N}$ and $B \in \mathbb{R}^{m \times m}$, their Kronecker product is denoted as $A \otimes B \in \mathbb{R}^{Nm \times Nm}$. For nonempty convex set $X \subseteq \mathbb{R}^n$, $T_X(\bar{x}) = \{\bar{a} \in \mathbb{R}^n : \bar{a} = \alpha(x - \bar{x}), x \in X, \alpha > 0\}$ denotes the tangent cone to X at $\bar{x}$, $N_X(\bar{x}) = \{\bar{b} \in \mathbb{R}^n : \langle \bar{b}, x - \bar{x} \rangle \leq 0, x \in X\}$ denotes the normal cone to X at $\bar{x}$.

II. PRELIMINARIES

A. Arithmetic of Intervals

We use $\mathcal{I}_{\mathbb{R}}$ to represent the class of all bounded closed intervals in $\mathbb{R}$, namely, $\mathcal{I}_{\mathbb{R}}$ can be defined by $\mathcal{I}_{\mathbb{R}} = \{[a^L, a^U] : a^L, a^U \in \mathbb{R} \text{ and } a^L < a^U\}$, in which a^L, a^U denote the lower and upper bounds of closed interval $[a^L, a^U]$, respectively. For any two closed intervals $a = [a^L, a^U], b = [b^L, b^U] \in \mathcal{I}_{\mathbb{R}}$, a partial order relation is usually employed to rank them, i.e., $a \preceq b$ corresponds to $a^L \leq b^L \wedge a^U \leq b^U$, $a \prec b$ corresponds to $a^L < b^L \wedge a^U \leq b^U$ (or $a^L \leq b^L \wedge a^U < b^U$, or $a^L < b^L \wedge a^U < b^U$). Moreover, $a + b = [a^L + b^L, a^U + b^U]$, $-a = [-a^U, -a^L]$, $a - b = a + (-b) = [a^L - b^U, a^U - b^L]$, $a + c = [a^L + c, a^U + c]$, $\forall c \in \mathbb{R}$, $ca = [ca^L, ca^U]$, $\forall c \geq 0$, and $ca = [ca^U, ca^L]$, $\forall c < 0$. Note that a singleton set $\{u\} \subseteq \mathbb{R}$ can be described as the interval $[u, u]$, such as $0 = [0, 0]$. Stefanini and Bede [39] introduced the generalized-Hukuhara difference (gH -difference) between two intervals as follows:

$$a \ominus_{\mathcal{H}} b = d \Leftrightarrow \begin{cases} a = b + d \\ \text{or } b = a + (-1)d. \end{cases} \quad (1)$$

For any $x \in \mathbb{R}^n$, the directional generalized-Hukuhara derivative (gH -derivative) of the interval-valued function $f^{\mathcal{I}}$ at x is defined by $f^{\mathcal{I}'}(x) = \lim_{r \rightarrow 0^+} (1/r)[f^{\mathcal{I}}(x + rd) \ominus_{\mathcal{H}} f^{\mathcal{I}}(x)]$, which can be obtained from metric space $(\mathcal{I}_{\mathbb{R}}, \mathcal{H})$. The corresponding metric $\mathcal{H}$ is described by

$$\mathcal{H}(\mathcal{A}, \mathcal{B}) = \max\{\sup_{a \in \mathcal{A}} \mathbf{D}(a, \mathcal{B}), \sup_{b \in \mathcal{B}} \mathbf{D}(\mathcal{A}, b)\} \quad (2)$$

with $\mathbf{D}(a, b) = \inf_{b \in \mathcal{B}} \|a - b\|$. If $f^{\mathcal{I}'}(x)$ exists, $f^{\mathcal{I}}$ is said to be generalized-Hukuhara differentiable (gH -differentiable) at x . Let $f^{\mathcal{I}}(x) = [f^L(x), f^U(x)]$, where $f^L(x), f^U(x)$ are real-valued functions and satisfy $f^L(x) \leq f^U(x)$, then $f^{\mathcal{I}}$ is gH -differentiable at x if f^L and f^U are differentiable at x . Moreover, $f^{\mathcal{I}}$ is convex if and only if f^L and f^U are both convex functions. On the basis of gH -derivative, the gradient of $f^{\mathcal{I}}$ at x is defined by

$$\nabla_{\mathcal{H}} f^{\mathcal{I}}(x) = \left[\left(\frac{\partial f^{\mathcal{I}}}{\partial x_1} \right)_{\mathcal{H}}(x), \dots, \left(\frac{\partial f^{\mathcal{I}}}{\partial x_n} \right)_{\mathcal{H}}(x) \right]^T \quad (3)$$

where $(\partial f^{\mathcal{I}} / \partial x_i)_{\mathcal{H}}(x), i \in \{1, \dots, n\}$ represents the i th partial gH -derivative of f at x , and meanwhile, $(\partial f^{\mathcal{I}} / \partial x_i)_{\mathcal{H}}(x) = [\min\{\partial f^L / \partial x_i(x), \partial f^U / \partial x_i(x)\}, \max\{\partial f^L / \partial x_i(x), (\partial f^U / \partial x_i)(x)\}]$, which is a closed interval. For the case where the interval-valued function $f^{\mathcal{I}} : X \subseteq \mathbb{R}^n \mapsto \mathcal{I}_{\mathbb{R}}$ is not gH -differentiable at x , its generalized-Hukuhara subdifferential (gH -subdifferential) of $f^{\mathcal{I}}$ at x can be described by $\partial_{\mathcal{H}} f^{\mathcal{I}}(x) = \{\Upsilon \in \mathcal{I}_{\mathbb{R}^n} : (x' - x)^T \Upsilon \preceq f^{\mathcal{I}}(x') \ominus_{\mathcal{H}} f^{\mathcal{I}}(x)\}$, where Υ denotes a generalized-Hukuhara sugradient (gH -sugradient), and besides, if $f^{\mathcal{I}}$ is gH -differentiable at $x \in X$ then $\nabla_{\mathcal{H}} f^{\mathcal{I}}(x) = \partial_{\mathcal{H}} f^{\mathcal{I}}(x)$.

B. Graph Theory

We use $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathcal{A})$ to denote a weighted undirected graph, in which $\mathcal{V} = (v_1, v_2, \dots, v_N)$ represents the set of vertices, $\mathcal{E} \subseteq \mathcal{V} \times \mathcal{V}$ represents the set of edges, $\mathcal{A} = [a_{ij}]_{N \times N} \in \mathbb{R}^{N \times N}$ represents the weighted adjacency matrix, where a_{ij} denotes the connection weight between the i th vertex and the j th vertex. Denote (v_i, v_j) as an edge from vertex v_i to vertex v_j . Graph $\mathcal{G}$ is undirected if $(v_i, v_j) \in \mathcal{E}$ corresponds to $(v_j, v_i) \in \mathcal{E}$. We consider undirected graphs with positive weights in this article, that is, $a_{ij} = a_{ji} > 0$ for $(v_i, v_j) \in \mathcal{E}$; if not, $a_{ij} = a_{ji} = 0$. Besides, $a_{ii} = 0$ which indicates undirected graph $\mathcal{G}$ have no self-connection. Define the degree matrix of $\mathcal{G}$ as $\mathcal{D}(\mathcal{G}) = \text{diag}\{\mathcal{D}_1, \mathcal{D}_2, \dots, \mathcal{D}_N\}$, in which $\mathcal{D}_i = \sum_{j=1}^N a_{ij}$, and then the Laplacian matrix can be defined by $L(\mathcal{G}) = \mathcal{D}(\mathcal{G}) - \mathcal{A}$. Note that if graph $\mathcal{G}$ is connected, $L(\mathcal{G})$ has a simple zero eigenvalue and all the other eigenvalues are positive.

C. Differential Inclusion and Nonsmooth Analysis

Assume that $\Phi : x \rightarrow \Phi(x)$ is a set-valued map from $\mathbb{E}$ to $\mathbb{R}^n$, namely, there exists a nonempty closed set $\Phi(x) \subseteq \mathbb{R}^n$ for any $x \in \mathbb{E} \subseteq \mathbb{R}^n$. Consider a differential inclusion on $X \subseteq \mathbb{R}^n$ with initial point $x_0 \in X$ as follows:

$$\dot{x}(t) \in \Phi(x(t)), t \in [0, T], x(0) = x_0 \quad (4)$$

where $x : [0, T] \mapsto \mathbb{R}^n$ is said a state solution of differential inclusion (4) with initial point x_0 if x is absolutely continuous and satisfies (4) for almost everywhere $t \in [0, T]$.

For a nonempty closed convex set $X \subseteq \mathbb{R}^n$, the projection of any vector $y \in \mathbb{R}^n$ onto it can be defined as

$$\mathcal{P}_X[y] = \arg \min_{x \in X} \|y - x\| \quad (5)$$

which satisfies [40]: 1) $\|\mathcal{P}_X[y] - \mathcal{P}_X[x]\| \leq \|y - x\|, \forall y, x \in \mathbb{R}^n$; 2) $\langle \mathcal{P}_X[y] - y, x - \mathcal{P}_X[y] \rangle \geq 0, \forall y \in \mathbb{R}^n, x \in X$; and 3) $\langle \mathcal{P}_X[y] - y, x - y \rangle \geq \|\mathcal{P}_X[y] - y\|^2, \forall y \in \mathbb{R}^n, x \in X$.

Definition 1 [28]: A function $f : \mathbb{R}^n \mapsto \mathbb{R}$ is said to be locally Lipschitz near x if there exists $\kappa > 0$ such that $\|f(y) - f(z)\| \leq \kappa\|y - z\|, \forall y, z \in \mathbb{B}(x, \sigma)$, where κ is the Lipschitz constant and $\mathbb{B}(x, r) = \{x' \in \mathbb{R}^n : \|x - x'\| < r\}$. The function f is said to be locally Lipschitz continuous on $\mathbb{R}^n$ if it is locally Lipschitz near any $x \in \mathbb{R}^n$.

Definition 2 [40]: A set-valued map $\Phi : \mathbb{E} \rightrightarrows \mathbb{R}^n$ is called upper semicontinuous at $x \in \mathbb{E}$ if for any $\sigma > 0$ there exist $\rho > 0$ such that $x' \in \mathbb{B}(x, \rho) \subseteq \mathbb{E}$ and $\Phi(x') \subseteq \mathbb{B}(\Phi(x), \sigma)$. The set-valued map Φ is upper semicontinuous on $\mathbb{E}$ if it is upper semicontinuous at any $x \in \mathbb{E}$.

More content on the above knowledge is available in [1], [28], [39], [40], and [41].

III. PROBLEM FORMULATION

We consider a distributed interval-valued convex optimization problem with general constraints over a network with N vertices, where the general constraints include local bound constraints and equality and inequality constraints. Each vertex has its own objective function and corresponding constraints, while the global objective function is the sum of N local interval-valued objective functions. Such an optimization problem can be described as

$$\begin{aligned} \min \quad & \sum_{i=1}^N f_i^{\mathcal{I}}(x) \\ \text{s.t.} \quad & A_i x = b_i, \quad g_i(x) \leq 0 \\ & x \in X_0 \triangleq \bigcap_{i=1}^N X_i, \quad i \in \{1, 2, \dots, N\} \end{aligned} \quad (6)$$

where $f_i^{\mathcal{I}}(x) = [f_i^L(x), f_i^U(x)] : \mathbb{R}^n \mapsto \mathcal{I}_{\mathbb{R}}$ is the local interval-valued objective function of the i th vertex, $x \in \mathbb{R}^n$ denotes the corresponding decision variable. $A_i \in \mathbb{R}^{m_i \times n}$, $b_i \in \mathbb{R}^{m_i}$, and $g_i = [g_{i1}, g_{i2}, \dots, g_{i\mathcal{J}(i)}]^T : \mathbb{R}^n \mapsto \mathbb{R}^{\mathcal{J}(i)}$ represents the vector composed of $\mathcal{J}(i)$ inequality constraint function values of the i th vertex. For any $i \in \{1, 2, \dots, N\}$ and $\iota \in \{1, 2, \dots, \mathcal{J}(i)\}$, $f_i^{\mathcal{I}}$ is convex but not necessarily gH -differentiable, $g_{i\iota}$ is a differentiable continuous convex function, and X_i is a closed convex set. Each vertex just knows its own local objective function and constraint information. To realize global optimization, the state outputs of all vertices are required to be controlled cooperatively.

The following assumptions are necessary to solve distributed interval-valued optimization problem (6).

Assumption 1: Graph $\mathcal{G}$ about vertex network is undirected and connected.

Assumption 2: For any $i \in \{1, 2, \dots, N\}$, the set $\{x \in \mathbb{R}^n : g_i(x) \leq 0\}$ is bounded.

Assumption 3: For any $x \in \mathbb{R}^n$, the intersection of $X_i, \text{int}\{x : g_i(x) \leq 0\}$, and $\text{int}\{x : A_i x = b_i\}$ is nonempty.

Under Assumption 1, distributed optimization problem (6) can be transformed into the following form:

$$\begin{aligned} \min \quad & [\mathbf{f}^L(\mathbf{x}), \mathbf{f}^U(\mathbf{x})] = \sum_{i=1}^N [f_i^L(x_i), f_i^U(x_i)] \\ \text{s.t.} \quad & A\mathbf{x} = \mathbf{b}, \quad g(\mathbf{x}) \leq 0 \\ & \mathbf{x} \in X \triangleq \prod_{i=1}^N X_i, \quad L\mathbf{x} = \mathbf{0} \end{aligned} \quad (7)$$

where $\mathbf{x} = [x_i]_{i=1}^N \in \mathbb{R}^{Nn}$, $A = \text{diag}\{A_1, A_2, \dots, A_N\} \in \mathbb{R}^{m \times Nn}$ denotes the block diagonal matrix generated by $A_1, A_2, \dots, A_N$, and $m = \sum_{i=1}^N m_i$. $\mathbf{b} = [b_1, b_2, \dots, b_m]^T \in \mathbb{R}^m$, $g(\mathbf{x}) = [g_i(x_i)]_{i=1}^N \in \mathbb{R}^{\mathcal{J}}$ with $\mathcal{J} = \sum_{i=1}^N \mathcal{J}(i)$. $\prod_{i=1}^N X_i$ denotes the Cartesian product generated by $X_1, X_2, \dots, X_N$. $L = L(\mathcal{G}) \otimes I_n \in \mathbb{R}^{Nn \times Nn}$, in which $L(\mathcal{G})$ is the Laplacian matrix associated with the undirected graph $\mathcal{G}$, I_n denotes identity matrix in $\mathbb{R}^{n \times n}$.

Remark 1: In general, when the communication graph of the vertices is not connected, problems (6) and (7) cannot be converted to each other. For the case of undirected connected graph $\mathcal{G}$, $L\mathbf{x} = \mathbf{0}$ means that the decision variables corresponding to each vertex achieve consensus, that is, $x_1 = x_2 = \dots = x_N$ [13, Lemma 3.1].

The following notations are used to facilitate later description. Let $\mathcal{A} = \{\mathbf{x} \in \mathbb{R}^{Nn} : A\mathbf{x} = \mathbf{b}\}$, $\mathcal{G} = \{\mathbf{x} \in \mathbb{R}^{Nn} : g(\mathbf{x}) \leq \mathbf{0}\}$, $\mathcal{L} = \{\mathbf{x} \in \mathbb{R}^{Nn} : L\mathbf{x} = \mathbf{0}\}$, and we use $\mathcal{F} = \mathcal{A} \cap \mathcal{G} \cap \mathcal{L} \cap X$ to represent the feasible region of optimization problem (7).

Definition 3 [15], [20]: A point $\mathbf{x}^*$ is said to be an efficient solution of the interval-valued optimization problem (7) if $[\mathbf{f}^L(\mathbf{x}), \mathbf{f}^U(\mathbf{x})] \not\prec [\mathbf{f}^L(\mathbf{x}^*), \mathbf{f}^U(\mathbf{x}^*)]$ for all $\mathbf{x} \in \mathcal{F}$.

Remark 2: In an optimization problem with the interval-valued objective function $[\mathbf{f}^L(\mathbf{x}), \mathbf{f}^U(\mathbf{x})]$, the lower bound $\mathbf{f}^L(\mathbf{x})$ and the upper bound $\mathbf{f}^U(\mathbf{x})$ are not arbitrary limits. They represent the best-case and worst case possible values of the objective function, given the inherent uncertainty in the problem's parameters.

Remark 3: Assumption 3 holds, which means that $X \cap \text{int}(\mathcal{G}) \cap \text{int}(\mathcal{A}) \neq \emptyset$. In fact, Assumption 3 can be viewed as an extension of the Slater's constraint qualification in the context of interval-valued convex optimization, which also provides a guarantee for the existence of solutions to the optimization problem (6). This assumption is mainly used to assist the proof of the efficiency condition about the distributed interval-valued optimization problem (6) (or (7)) in Section IV.

Assumption 2 holds. That is to say there exists $r > 0$ such that $\{x \in \mathbb{R}^n : g_i(x) \leq 0\} \subseteq \mathbb{B}(0, r)$. We denote

$$\begin{aligned} p_{i\iota}(x_i) &= g_{i\iota}(x_i) + \max\{\|x_i\|, r\} - r \\ p_i(x_i) &= \sum_{\iota=1}^{\mathcal{J}(i)} \max\{p_{i\iota}(x_i), 0\} \\ p(\mathbf{x}) &= [p_1(x_1), p_2(x_2), \dots, p_N(x_N)]^T \end{aligned}$$

and define a penalty function $\Gamma(\mathbf{x}) \triangleq \sum_{i=1}^N p_i(x_i)$, then $\Gamma(\mathbf{x})$ is a convex function and satisfies [33]: 1) $\Gamma(\mathbf{x})$ is coercive,

i.e., $\lim_{\|x\| \rightarrow \infty} \Gamma(x) = \infty$; and 2) the subdifferential of Γ at x can be described as

$$\partial\Gamma(x) = \begin{cases} \{\mathbf{0}\}, & x \in \text{int}(\mathcal{G}) \\ \sum_{i \in P_0} [0, 1] u_i \otimes \partial p_i(x_i), & x \in \text{bd}(\mathcal{G}) \\ \sum_{i \in P_+} u_i \otimes \partial p_i(x_i) \\ \quad + \sum_{i \in P_0} [0, 1] u_i \otimes \partial p_i(x_i), & x \notin \mathcal{G} \end{cases}$$

where $u_i = \overbrace{[0, \dots, 1, \dots, 0]}^N$ represents the N -dimensional unit vector in which all terms are 0 except the i th term is 1, $P_0 = \{i \in \{1, 2, \dots, N\} : p_i(x_i) = 0\}$, and $P_+ = \{i \in \{1, 2, \dots, N\} : p_i(x_i) > 0\}$; and 3) $\{x \in \mathbb{R}^{Nn} : p(x) \leq \mathbf{0}\} = \{x \in \mathbb{R}^{Nn} : g(x) \leq \mathbf{0}\} = \mathcal{G}$.

IV. ALGORITHM DESIGN

According to the discussion in the previous section, a projection neurodynamic approach involving gH -subgradient is proposed to solve distributed interval-valued optimization problem (7) in this section. The dynamic model can be described by the following differential inclusion:

$$\begin{cases} \dot{x}_i \in -x_i + \mathcal{P}_{X_i} \left[x_i - \delta_i(x_i) \partial_{\mathcal{H}} F_i(x_i) - \partial p_i(x_i) \right. \\ \quad \left. - \partial \|A_i x_i - b_i\|_1 - \sum_{j \in \mathcal{N}_i} a_{ij}(x_i - x_j) \right. \\ \quad \left. - \frac{1}{2} \sum_{j \in \mathcal{N}_i} a_{ij}(v_i - v_j) \right], \\ \dot{v}_i = x_i \end{cases} \quad (8a)$$

in which $x_i \in \mathbb{R}^n$ represents the i -neuron's state, a_{ij} displays the connection relationship between the i th neuron and the j th neuron, $\mathcal{N}_i$ is the set consisting of the i th neuron's neighbors, and $\vartheta_i, \vartheta_j \in \mathbb{R}^n$ can be viewed as the i, j th neurons' hidden states. The hidden states of neurons also achieve information interaction through connection relationships. We define $\delta_i(x_i)$ with a set-valued limiter function $\phi : \mathbb{R} \rightarrow [0, 1]$, and specifically, ϕ is derived by a piecewise function

$$\phi(\star) = \begin{cases} 1, & \star > 0 \\ [0, 1], & \star = 0 \\ 0, & \star < 0 \end{cases}$$

and $\delta_i(x_i) = 1 - \phi(\mathcal{P}_i(x_i))$ where $\mathcal{P}_i(x_i) = \sum_{i=1}^{\mathcal{I}(i)} g_{il}(x_i) + \|A_i x_i - b_i\|_1$. Note that $\partial \|A_i x_i - b_i\|_1 = A_i^T \theta(A_i x_i - b_i)$ and $\theta(A_i x_i - b_i) = [\theta_{\bullet}(\tau_{i,1}), \theta_{\bullet}(\tau_{i,2}), \dots, \theta_{\bullet}(\tau_{i,m_i})]^T \in \mathbb{R}^{m_i}$, where

$$\theta_{\bullet}(\tau_{i,\kappa}) = \begin{cases} 1, & \tau_{i,\kappa} > 0 \\ [-1, 1], & \tau_{i,\kappa} = 0 \\ -1, & \tau_{i,\kappa} < 0. \end{cases}$$

In dynamic inclusion (8a), $\partial_{\mathcal{H}} F_i(x_i)$ represents the gH -subdifferential of $f_i^{\mathcal{I}}$ at x_i , and $\partial_{\mathcal{H}} F_i(x_i) \ni$

$\varepsilon_1 \partial f_i^{\mathcal{I}}(x_i) + \varepsilon_2 \partial f_i^{\mathcal{U}}(x_i)$ with $\varepsilon_1, \varepsilon_2 \in [0, 1]$ and $\varepsilon_1 + \varepsilon_2 = 1$. Denote by $\gamma_{i\mathcal{I}} \in \partial_{\mathcal{H}} F_i(x_i)$ a gH -subgradient, and it satisfies $(x'_i - x_i)^T \gamma_{i\mathcal{I}} \preceq f_i^{\mathcal{I}}(x'_i) \ominus_{\mathcal{H}} f_i^{\mathcal{I}}(x_i)$. Moreover, $\partial_{\mathcal{H}} F_i(x_i)$ can also be written in the interval-valued vector form

$$\partial_{\mathcal{H}} F_i(x_i) = [(\partial_{x_{i,1}} f_i^{\mathcal{I}})_{\mathcal{H}}(x_i), \dots, (\partial_{x_{i,n}} f_i^{\mathcal{I}})_{\mathcal{H}}(x_i)]^T$$

in which $(\partial_{x_{i,l}} f_i^{\mathcal{I}})_{\mathcal{H}}(x_i)$ represents the l th generalized partial gH -derivative of $f_i^{\mathcal{I}}$ at x_i . For the case where $f_i^{\mathcal{I}}$ is gH -differentiable at x_i , the gH -subdifferential $\partial_{\mathcal{H}} F_i(x_i)$ is reduced to the gH -gradient, i.e., $\partial_{\mathcal{H}} F_i(x_i) = \nabla_{\mathcal{H}} F_i(x_i) \triangleq [(\partial f_i^{\mathcal{I}} / \partial x_{i,1})_{\mathcal{H}}(x_i), \dots, (\partial f_i^{\mathcal{I}} / \partial x_{i,n})_{\mathcal{H}}(x_i)]^T$, where $(\partial f_i^{\mathcal{I}} / \partial x_{i,l})_{\mathcal{H}}(x_i)$ denotes the l th partial gH -derivative of $f_i^{\mathcal{I}}$ at x_i .

Remark 4: Since g_{il} is a convex function and $f_i^{\mathcal{I}}$ is convex in the sense of interval-valued functions, then $f_i^{\mathcal{I}}$ and g_{il} satisfy the local Lipschitz condition. Meanwhile, the set-valued maps $\partial_{\mathcal{H}} f_i^{\mathcal{I}}$ and ∂g_{il} are upper semicontinuous, and the corresponding sets $\partial_{\mathcal{H}} f_i^{\mathcal{I}}(x_i)$ and $\partial g_{il}(x_i)$ are nonempty, compact and convex. Analogously, the set-valued maps ∂p_i and $\partial \|A_i x_i - b_i\|_1$ are upper semicontinuous with nonempty, compact, and convex values. Therefore, the right-hand side of differential inclusion (8a) is an upper semicontinuous set-valued map, and the set derived from it is a nonempty, compact, and convex. With the analogy results of Aubin and Cellina based on Peano's theorem [40, Lemma 3.1], we can see that the existence of the state solution of differential inclusion (8a) in the sense of Filippov can be guaranteed.

Lemma 1: For any initial condition $x_{i0} \in \mathbb{R}^n$, when the right-hand side of differential inclusion (8a) satisfies a one-sided Lipschitz condition, the Filippov solution is continuous and unique.

Proof: Suppose x_i and x'_i are two solutions of differential inclusion (8a) starting from the same initial condition x_{i0} . Consider the evolution of the squared distance between them: $(1/2)(d/dt)\|x_i - x'_i\|^2 = \langle \dot{x}_i - \dot{x}'_i, x_i - x'_i \rangle$. By the one-sided Lipschitz condition, we have $\langle \dot{x}_i - \dot{x}'_i, x_i - x'_i \rangle \leq \kappa \|x_i - x'_i\|^2$ with κ being the Lipschitz constant, which yields $(d/dt)\|x_i - x'_i\|^2 \leq 2\kappa \|x_i - x'_i\|^2$. Since $\|x_{i0} - x'_{i0}\|^2 = 0$, Grönwall's inequality implies that $\|x_i - x'_i\|^2 = 0$ for all $t \geq 0$, the solution of differential inclusion (8a) is continuous and unique. ■

With the notations given in the previous section, neurodynamics (8) can be rewritten in the following compact forms:

$$\begin{cases} \dot{x} \in -x + \mathcal{P}_X \left[x - \delta \partial_{\mathcal{H}} F(x) - \partial \Gamma(x) - \partial \|Ax - b\|_1 \right. \\ \quad \left. - Lx - \frac{1}{2} L\vartheta \right] \\ \dot{\vartheta} = x \end{cases} \quad (9a)$$

in which $\partial_{\mathcal{H}} F(x) = [\delta_i(x_i) \partial_{\mathcal{H}} F_i(x_i)]_{i=1}^N$ and $\partial \|Ax - b\|_1 = A^T [\theta(A_i x_i - b_i)]_{i=1}^N$. Similar to the structure of the state variable x , $\vartheta = [\vartheta_i]_{i=1}^N \in \mathbb{R}^{Nn}$.

Remark 5: In the construction of neurodynamics, the authors [30], [32] use operator $I_{Nn} - A^T(AA^T)^{-1}A$ to handle equality constraints with the highest computational complexity $\mathcal{O}(n^3)$, where $I_{Nn} \in \mathbb{R}^{Nn \times Nn}$ is the identity matrix, $A^T(AA^T)^{-1}A$ is the orthogonal projection matrix. To some extent, algorithm (9) can reduce the calculation load because it avoids the calculation

of inverse matrix. The main function of the projection operator $\mathcal{P}_X[\cdot]$ is to push the state solution toward the constraint set X . $\mathcal{P}_X[\cdot]$ processes the box constraints in a component-by-component clipping manner, or handles the ball constraints in a normalized way, with a computational complexity $\mathcal{O}(n)$. For X to be a simplex, the complexity of $\mathcal{P}_X[\cdot]$ is $\mathcal{O}(n \log n)$. Affine type $Ax = b$ is discussed separately from X , because the computational complexity of $\mathcal{P}_X[\cdot]$ for the affine bundle is approximately $\mathcal{O}(n^3)$. Unlike the previous Lagrange multiplier methods [10], [35], [36] with a computational complexity of $\mathcal{O}(n)$, $\mathcal{P}_X[\cdot]$ does not increase the number of neurons in the input layer. Compared with another projection method $\mathcal{P}_{T_X(x)}[\cdot]$ mentioned in [10], $\mathcal{P}_X[\cdot]$ avoids the calculation of tangent cone.

Theorem 1: The set X is a positively invariant region for the state solution $x(t)$ of neurodynamic system (9a) with the initial state $x_0 \in X$. Alternatively, the state solution $x(t)$ converges globally and exponentially to set X .

Proof: For any initial state $x_0 \in \mathbb{R}^{Nn}$, there exists $\ell^{\mathcal{I}}(t) = [\ell^L(t), \ell^U(t)] \in \mathcal{P}_X[x - \partial_{\mathcal{F}}^{\delta} \mathcal{F}(x) - \partial \Gamma(x) - \partial \|Ax - b\|_1 - Lx - (1/2)L\vartheta]$ such that $\dot{x} \in -x + \ell^{\mathcal{I}}(t)$. Further, we know that there exist $\varepsilon_1, \varepsilon_2 \in [0, 1]$ such that $\dot{x} = -x + \varepsilon_1 \ell^L(t) + \varepsilon_2 \ell^U(t)$ where $\varepsilon_1 + \varepsilon_2 = 1$. Denote $q(t) = e^t x$ and take the derivative of it, we have the following:

$$\dot{q}(t) = e^t x + e^t \dot{x} = e^t (x + \dot{x}) = e^t (\varepsilon_1 \ell^L(t) + \varepsilon_2 \ell^U(t)) \quad (10)$$

it follows that $\int_0^t \dot{q}(\varrho) d\varrho = \int_0^t e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho)) d\varrho$. Accordingly, $q(t) - q(0) = \int_0^t e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho)) d\varrho$, and then $q(t) = q(0) + \int_0^t e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho)) d\varrho = x_0 + \int_0^t e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho)) d\varrho$. It obtains from $q(t) = e^t x$ that $e^t x = x_0 + \int_0^t e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho)) d\varrho$, which implies

$$\begin{aligned} x &= e^{-t} x_0 + e^{-t} \int_0^t e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho)) d\varrho \\ &= e^{-t} x_0 + (1 - e^{-t}) \int_0^t \frac{e^{\varrho} (\varepsilon_1 \ell^L(\varrho) + \varepsilon_2 \ell^U(\varrho))}{e^t - 1} d\varrho \\ &= e^{-t} x_0 + (1 - e^{-t}) (\varepsilon_1 \ell^L(\tilde{\varrho}) + \varepsilon_2 \ell^U(\tilde{\varrho})) \int_0^t \frac{e^{\varrho}}{e^t - 1} d(\varrho) \\ &= e^{-t} x_0 + (1 - e^{-t}) (\varepsilon_1 \ell^L(\tilde{\varrho}) + \varepsilon_2 \ell^U(\tilde{\varrho})) \\ &\in e^{-t} x_0 + (1 - e^{-t}) \ell^{\mathcal{I}}(\tilde{\varrho}) \end{aligned}$$

where $\tilde{\varrho} \in (0, t)$. If $x_0 \in X$, then $x \in X$ for $t \in [0, \infty)$, which means X is a positively invariant set of neurodynamic system (9a). Conversely, if $x_0 \notin X$, then the state solution $x(t)$ converges globally and exponentially to set X . ■

The projection operator $\mathcal{P}_X[\cdot]$ pushes the state solution of neurodynamics (9) to the constraint set X via a relatively fast convergence rate, while ensuring that it does not exceed this region. The following is the efficiency condition of the solution to distributed interval-valued convex optimization problem (7) based on the analysis of the designed neurodynamic structure.

Theorem 2: Assumptions 2 and 3 hold. x^* is an efficient solution to interval-valued optimization problem (7) if and only

if there exist $\vartheta^* \in \mathbb{R}^{Nn}$, $\gamma_{\mathcal{I}}^* \in \partial_{\mathcal{F}}^{\delta} \mathcal{F}(x^*)$, $\zeta^* \in \partial \Gamma(x^*)$, $\mu^* \in \partial \|Ax^* - b\|_1$ such that

$$\begin{cases} 0 \in -x^* + \mathcal{P}_X \left[x^* - \hat{\gamma}_{\mathcal{I}}^* - \zeta^* - \mu^* - Lx^* - \frac{1}{2} L\vartheta^* \right], & (11a) \\ g(x^*) \leq 0, \quad Ax^* = b, & (11b) \\ Lx^* = 0, \quad x^* \in X. & (11c) \end{cases}$$

Proof: According to Ruzschiński's explanation of the optimality condition for constrained convex optimization problems [41, Th. 3.33], it can be seen that x^* is an efficient solution of problem (7) if and only if

$$\begin{cases} 0 \in \partial_{\mathcal{F}}^{\delta} F(x^*) + \frac{1}{2} L\vartheta^* + N_{\mathcal{F}}(x^*) & (12a) \\ x^* \in \mathcal{F} & (12b) \end{cases}$$

where (12b) and (11b)–(11c) are equivalent, and what needs to be proved is the equivalence between (12a) and (11a). ■

We first explain the relationship between $N_{\mathcal{F}}(x^*)$, $N_{\mathcal{A}}(x^*)$, $N_{\mathcal{G}}(x^*)$, and $N_X(x^*)$. From [41, Lemma 2.40], we have $N_{\mathcal{F}}(x^*) = N_{\mathcal{A} \cap \mathcal{G} \cap \mathcal{L} \cap X}(x^*) = N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) + N_{\mathcal{L}}(x^*) + N_X(x^*)$ for any $x^* \in \mathcal{F}$, which implies $N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) \subseteq N_{\mathcal{F}}(x^*)$ where $N_{\mathcal{G}}(x^*) = \bigcup_{\nu \geq 0} \nu \partial \Gamma(x^*)$. When $x \in \mathcal{A}$, one has $Ax - b = 0$, which yields $\langle A^T \eta, x - x^* \rangle = \eta^T (Ax - Ax^*) = \eta^T (Ax - b) = \eta^T (b - b) = 0$ for any $\eta \in \mathbb{R}^m$, it follows from the geometric concept of normal cone that $\{A^T \eta : \eta \in \mathbb{R}^m\} \subseteq N_{\mathcal{A}}(x^*)$. In turn, for any $\rho \in N_{\mathcal{A}}(x^*)$ we have $\rho^T (x - x^*) \leq 0$ and there exists $\eta^* \in \{[\theta(A_i x_i^* - b_i)]_{i=1}^N\}$ such that $\rho = -A^T \eta^* \in \{A^T \eta : \eta \in \mathbb{R}^m\}$, where x_i^* denotes the local component of efficient solution, i.e., $x^* = [x_i^*]_{i=1}^N$. Then, $N_{\mathcal{A}}(x^*) \subseteq \{A^T \eta : \eta \in \mathbb{R}^m\}$, it follows that $N_{\mathcal{A}}(x^*) = \{A^T \eta : \eta \in \mathbb{R}^m\}$. On the other hand, we have $\bigcup_{\nu \geq 0} \nu \partial \|Ax^* - b\|_1 = \bigcup_{\nu \geq 0} \nu \{A^T [\theta(A_i x_i^* - b_i)]_{i=1}^N\} = \{A^T \eta : \eta \in \mathbb{R}^m\}$, which suggests $N_{\mathcal{A}}(x^*) = \bigcup_{\nu \geq 0} \nu \partial \|Ax^* - b\|_1$. We know $N_{\mathcal{L}}(x^*) = \bigcup_{\nu \geq 0} \nu \{Lx^* : Lx^* = 0\}$ by a similar derivation. By combining $N_{\mathcal{L}}(x^*) = \{0\}$, $\{0\} \subseteq \bigcup_{\nu \geq 0} \nu \partial \|Ax^* - b\|_1$ and $\{0\} \subseteq \bigcup_{\nu \geq 0} \nu \partial \Gamma(x^*)$, one obtains $N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) + N_{\mathcal{L}}(x^*) = N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*)$. Further, we get $N_{\mathcal{F}}(x^*) = N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) + N_X(x^*)$.

In what follows, we prove that (12a) $\Rightarrow$ (11a). Suppose that (12a) holds, i.e.,

$$0 \in \partial_{\mathcal{F}}^{\delta} \mathcal{F}(x^*) + \frac{1}{2} L\vartheta^* + N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) + N_X(x^*) \quad (13)$$

there must exist $\gamma_{\mathcal{I}}'^* \in \partial_{\mathcal{F}}^{\delta} \mathcal{F}(x^*)$, $\zeta'^* \in \partial \Gamma(x^*)$, and $\mu'^* \in \partial \|Ax^* - b\|_1$ such that $-\gamma_{\mathcal{I}}'^* - (1/2)L\vartheta^* - \zeta'^* - \mu'^* \subseteq N_X(x^*)$. We analyze and discuss $N_X(x^*)$ in the following two cases.

Case 1: If $x^* \in \text{int}(X)$ then $T_X(x^*) = \mathbb{R}^{Nn}$, which suggests $N_X(x^*) = \{0\}$, it obtains that $-\gamma_{\mathcal{I}}'^* - (1/2)L\vartheta^* - \zeta'^* - \mu'^* = 0$. Replace $\gamma_{\mathcal{I}}^*$, ζ^* , and μ^* in the right-hand side of (11a) with $\gamma_{\mathcal{I}}'^*$, ζ'^* , and μ'^* , we have $-x^* + \mathcal{P}_X[x^*] = -x^* + x^* = 0$. (11a) holds.

Case 2: If $x^* \in \text{bd}(X)$, there exists a corresponding $-\hat{h}_{\mathcal{I}}'^* \subseteq N_X(x^*)$ such that $-\gamma_{\mathcal{I}}'^* - (1/2)L\vartheta^* - \zeta'^* - \mu'^* = -\hat{h}_{\mathcal{I}}'^*$, where $-\hat{h}_{\mathcal{I}}'^* \neq 0$. Since $x^* - \hat{h}_{\mathcal{I}}'^* \in N_X(x^*)$, then

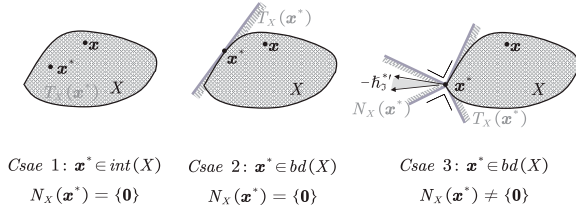


Fig. 1. Normal cones and efficient solution locations.

$\mathcal{P}_X[x^* - \bar{h}_T'] = x^*$. Further, $-x^* + \mathcal{P}_X[x^* - \bar{h}_T'] = 0$. (11a) holds.

(11a) $\Rightarrow$ (12a). Conversely, suppose that (11a) holds, we have $x^* \in \mathcal{P}_X[x^* - \gamma_T^* - \zeta^* - \mu^* - Lx^* - (1/2)L\vartheta^*]$, then $0 \in \gamma_T^* + \zeta^* + \mu^* + (1/2)L\vartheta^*$, which yields $0 \in \partial_{\mathcal{F}}^\delta \mathcal{F}(x^*) + (1/2)L\vartheta^* + N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) \subseteq \partial_{\mathcal{F}}^\delta \mathcal{F}(x^*) + (1/2)L\vartheta^* + N_{\mathcal{F}}(x^*)$. Hence, (12a) holds. The proof is complete.

Remark 6: To better address nonsmooth optimization problems, $\|Ax - b\|_1$ is used as a penalty function to eliminate equality constraints, whose subdifferential corresponds to a set of continuous projection functions to ensure the convergence of algorithm (9). Concretely, the absolute value of each component of $[\theta(A_i x_i - b_i)]_{i=1}^N$ is equal to 1 if $x \notin \mathcal{A}$, which indicates that the state solution searches the equality constraint set via a faster rate. If only the quadratic penalty term $(1/2)\|Ax - b\|_2^2$ is introduced, it cannot effectively ensure the calculation accuracy of the solution to problem (7). This results in the need for a continuously increasing penalty parameter, which will lead to an increase in the condition number of the Hessian matrix.

Remark 7: The proof idea here is to turn problem (7) into a set constrained interval-valued optimization problem with the form $\min_{x \in \mathcal{F}} [f^L(x), f^U(x)]$, and according to the geometric interpretation of the cones, an efficiency conclusion of solutions of the interval-valued convex optimization is established around the normal cone-based optimality condition. Fig. 1 shows three cases of the normal cone associated with the efficient solution locations. When the efficient solution $x^* \in \text{int}(X)$ for problem (7), one gets $N_X(x^*) = \{0\}$, that is $N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) + N_X(x^*) = N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*)$, and thus $N_{\mathcal{A}}(x^*) + N_{\mathcal{G}}(x^*) = N_{\mathcal{F}}(x^*)$. For Case 3, $-\bar{h}_T'$ denotes the negative gH -subgradient of interval-valued function $[f^L(x), f^U(x)]$ at x^* .

V. CONVERGENCE ANALYSIS

The boundedness of state solution of neurodynamics is of great significance, which is the premise to verify the convergence and validity of algorithm. Before analyzing the convergence of algorithm (9), we first discuss the boundedness of state trajectory $(x, L\vartheta)$.

Theorem 3: For any given initial point (x_0, ϑ_0) , the state trajectory $(x, L\vartheta)$ of the neurodynamic algorithm (9) is bounded.

Proof: Let x^* be an efficient solution to the distributed interval-valued optimization problem (7), then there exists $\vartheta^* \in \mathbb{R}^{Nn}$ satisfying the differential inclusion and equation in (9).

Consider the following candidate interval-valued Lyapunov function:

$$V^{\mathcal{I}}(x, \vartheta) = w^{\mathcal{I}}(x, \vartheta) - w^{\mathcal{I}}(x^*, \vartheta^*) - \Lambda^{\mathcal{I}}(x, \vartheta, x^*, \vartheta^*) + \frac{1}{2}\|x - x^*\|^2 + \frac{1}{2} \left\langle \frac{1}{2}(\vartheta - \vartheta^*), \frac{1}{2}L(\vartheta - \vartheta^*) \right\rangle \quad (14)$$

in which $w^{\mathcal{I}}(x, \vartheta) = \int_{\mathcal{F}} \partial_{\mathcal{F}}^\delta \mathcal{F}(x) dx + \Gamma(x) + \|Ax - b\|_1 + (1/2)\langle x + (1/2)\vartheta, L(x + (1/2)\vartheta) \rangle$, $w^{\mathcal{I}}(x^*, \vartheta^*) = \int_{\mathcal{F}} \partial_{\mathcal{F}}^\delta \mathcal{F}(x^*) dx + \Gamma(x^*) + \|Ax^* - b\|_1 + (1/2)\langle x^* + (1/2)\vartheta^*, L(x^* + (1/2)\vartheta^*) \rangle$, with $\int_{\mathcal{F}} \partial_{\mathcal{F}}^\delta \mathcal{F}(x) dx$ and $\int_{\mathcal{F}} \partial_{\mathcal{F}}^\delta \mathcal{F}(x^*) dx$ represent the Lebesgue integral of $\partial_{\mathcal{F}}^\delta \mathcal{F}(x)$ and $\partial_{\mathcal{F}}^\delta \mathcal{F}(x^*)$ on the set $\mathcal{F}$, respectively. $\Lambda^{\mathcal{I}}(x, \vartheta, x^*, \vartheta^*) = \langle x - x^*, \gamma_T^* + \zeta^* + \mu^* + (1/2)L\vartheta^* \rangle + \langle (1/2)(\vartheta - \vartheta^*), (1/2)L\vartheta^* \rangle$, and $\gamma_T^* \in \partial_{\mathcal{F}}^\delta \mathcal{F}(x^*)$, $\zeta^* \in \partial \Gamma(x^*)$, $\mu^* \in \partial \|Ax^* - b\|_1$. Taking the generalized gH -derivative of $w^{\mathcal{I}}(x, \vartheta)$ and $\Lambda^{\mathcal{I}}(x, \vartheta, x^*, \vartheta^*)$ with respect to x, ϑ and it obtains that

$$\begin{aligned} \frac{dw^{\mathcal{I}}}{dx} &= \partial_{\mathcal{F}}^\delta \mathcal{F}(x^*) + \partial \Gamma(x) + \partial \|Ax - b\|_1 + L \left(x + \frac{1}{2}\vartheta \right) \\ \frac{dw^{\mathcal{I}}}{d\vartheta} &= \frac{1}{2}L \left(x + \frac{1}{2}\vartheta \right) \\ \frac{d\Lambda^{\mathcal{I}}}{dx} &= \gamma_T^* + \zeta^* + \mu^* + \frac{1}{2}L\vartheta^*, \quad \frac{d\Lambda^{\mathcal{I}}}{d\vartheta} = \frac{1}{4}L\vartheta^* \end{aligned}$$

then the generalized gH -derivative of $V^{\mathcal{I}}(x, \vartheta)$ along the state solution of neurodynamic system (9) can be expressed as follows:

$$\begin{aligned} \dot{V}^{\mathcal{I}}(x, \vartheta) &= \left(\frac{dw^{\mathcal{I}}}{dx} - \frac{d\Lambda^{\mathcal{I}}}{dx} + x - x^* \right)^T \frac{dx}{dt} \\ &\quad + \left(\frac{dw^{\mathcal{I}}}{d\vartheta} - \frac{d\Lambda^{\mathcal{I}}}{d\vartheta} + \frac{1}{4}L(\vartheta - \vartheta^*) \right)^T \frac{d\vartheta}{dt} \\ &= \left(\gamma_T + \zeta + \mu + L \left(x + \frac{1}{2}\vartheta \right) - \gamma_T^* - \zeta^* - \mu^* \right. \\ &\quad \left. - \frac{1}{2}L\vartheta^* + x - x^* \right)^T \frac{dx}{dt} + \left(\frac{1}{2}L \left(x + \frac{1}{2}\vartheta \right) \right. \\ &\quad \left. - \frac{1}{4}L\vartheta^* + \frac{1}{4}L(\vartheta - \vartheta^*) \right)^T \frac{d\vartheta}{dt}. \end{aligned}$$

Further, it follows by the arbitrariness of $\mu \in \partial \|Ax - b\|_1$, $\zeta \in \partial \Gamma(x)$, $\gamma_T \in \partial_{\mathcal{F}}^\delta \mathcal{F}(x)$ that

$$\begin{aligned} \dot{V}^{\mathcal{I}}(x, \vartheta) &\lesssim - \sup_{\gamma_T, \zeta, \mu} \left(\gamma_T + \zeta + \mu + L \left(x + \frac{1}{2}\vartheta \right) - \gamma_T^* - \zeta^* \right. \\ &\quad \left. - \mu^* - \frac{1}{2}L\vartheta^* + x - x^* \right)^T (-x + \mathcal{P}_X[x - \gamma_T - \zeta \\ &\quad - \mu - L \left(x + \frac{1}{2}\vartheta \right)]) + \left(\frac{1}{2}L(\vartheta - \vartheta^*) + \frac{1}{2}Lx \right)^T x \\ &= \sup_{h_T} (\bar{h}_T - \bar{h}_T^* + x - x^*)^T (-x + \mathcal{P}_X[x - \bar{h}_T]) \\ &\quad + \left(\frac{1}{2}L(\vartheta - \vartheta^*) + \frac{1}{2}Lx \right)^T x \quad (15) \end{aligned}$$

where $\bar{h}_T = \gamma_T + \zeta + \mu + L(x + (1/2)\vartheta)$, $\bar{h}_T^* = \gamma_T^* + \zeta^* + \mu^* + L(x^* + (1/2)\vartheta^*) = \gamma_T^* + \zeta^* + \mu^* + (1/2)L\vartheta^*$. Since

$\mathcal{G}$ is an undirected graph, L is a real symmetric matrix, i.e., $L^T = L$. We denote

$$q_{1\mathcal{I}} = (\mathbf{h}_{\mathcal{I}} - \mathbf{h}_{\mathcal{I}}^* + \mathbf{x} - \mathbf{x}^*)^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*) \quad (16)$$

$$q_{2\mathcal{I}} = (\mathbf{h}_{\mathcal{I}} - \mathbf{h}_{\mathcal{I}}^* + \mathbf{x} - \mathbf{x}^*)^T (\mathbf{x}^* - \mathbf{x}) + \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*)^T L\mathbf{x} + \frac{1}{2}\mathbf{x}^T L\mathbf{x} \quad (17)$$

it follows that

$$\dot{V}^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) \lesssim \sup_{q_{1\mathcal{I}}, q_{2\mathcal{I}}} (q_{1\mathcal{I}} + q_{2\mathcal{I}}). \quad (18)$$

By transforming $q_{1\mathcal{I}}$, we get the following:

$$\begin{aligned} q_{1\mathcal{I}} &= (\mathbf{h}_{\mathcal{I}} - \mathbf{h}_{\mathcal{I}}^* - \mathbf{x} + \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}])^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*) \\ &\quad + (\mathbf{x} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] + \mathbf{x} - \mathbf{x}^*)^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*) \\ &= \underbrace{-(\mathbf{x} - \mathbf{h}_{\mathcal{I}} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}])^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*)}_{q_{1,1\mathcal{I}}} \\ &\quad - \underbrace{(\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*)^T \mathbf{h}_{\mathcal{I}}^*}_{q_{1,2\mathcal{I}}} \\ &\quad + \underbrace{(\mathbf{x} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] + \mathbf{x} - \mathbf{x}^*)^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*)}_{q_{1,3\mathcal{I}}} \end{aligned} \quad (19)$$

where $q_{1,1\mathcal{I}} = (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - (\mathbf{x} - \mathbf{h}_{\mathcal{I}}))^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x}^*) \lesssim 0$. For $\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}], \mathbf{x}^* \in X$ and $-\mathbf{h}_{\mathcal{I}}^* \in N_X(\mathbf{x}^*)$, one gets $q_{1,2\mathcal{I}} \lesssim 0$ by the geometric concept of normal cone, which suggests $q_{1\mathcal{I}} \lesssim q_{1,3\mathcal{I}} = (\mathbf{x} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] + \mathbf{x} - \mathbf{x}^*)^T (\mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}] - \mathbf{x} + \mathbf{x} - \mathbf{x}^*) = -\|\mathbf{x} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}]\|^2 + \|\mathbf{x} - \mathbf{x}^*\|^2$. On the other hand, for $q_{2\mathcal{I}}$, we get the following:

$$\begin{aligned} q_{2\mathcal{I}} &= (\gamma_{\mathcal{I}} - \gamma_{\mathcal{I}}^*)^T (\mathbf{x}^* - \mathbf{x}) + (\zeta - \zeta^*)^T (\mathbf{x}^* - \mathbf{x}) \\ &\quad + (\mu - \mu^*)^T (\mathbf{x}^* - \mathbf{x}) + \left(L \left(\mathbf{x} + \frac{1}{2}\boldsymbol{\vartheta} \right) - \frac{1}{2}L\boldsymbol{\vartheta}^* \right)^T \\ &\quad \times (\mathbf{x}^* - \mathbf{x}) - \|\mathbf{x} - \mathbf{x}^*\|^2 + \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*)^T L\mathbf{x} + \frac{1}{2}\mathbf{x}^T L\mathbf{x} \\ &= (\gamma_{\mathcal{I}} - \gamma_{\mathcal{I}}^*)^T (\mathbf{x}^* - \mathbf{x}) - (\zeta - \zeta^*)^T (\mathbf{x}^* - \mathbf{x}) \\ &\quad + (\mu - \mu^*)^T (\mathbf{x}^* - \mathbf{x}) - \|\mathbf{x} - \mathbf{x}^*\|^2 - \frac{1}{2}\mathbf{x}^T L\mathbf{x}. \end{aligned} \quad (20)$$

Since $\gamma_{\mathcal{I}}^* \in \partial_{\mathcal{H}}^{\delta} \mathcal{F}(\mathbf{x}^*)$ and $\partial_{\mathcal{H}}^{\delta} \mathcal{F}(\mathbf{x}^*) = [\delta_i(x_i^*) \partial_{\mathcal{H}} \mathcal{F}_i(x_i^*)]_{i=1}^N \subseteq [0, 1][\partial_{\mathcal{H}} \mathcal{F}_i(x_i^*)]_{i=1}^N = [0, 1][[(\partial_{x_{i,t}} f_i^T)(x_i^*)]_{t=1}^n]_{i=1}^N$, it follows by the convexity of f_i^T in the sense of interval-valued function that $(\gamma_{\mathcal{I}} - \gamma_{\mathcal{I}}^*)^T (\mathbf{x}^* - \mathbf{x}) \lesssim 0$. Besides, from the convexity of $\Gamma(\mathbf{x})$ and $\|A\mathbf{x} - \mathbf{b}\|_1$ we have $(\zeta - \zeta^*)^T (\mathbf{x}^* - \mathbf{x}) \leq 0$ and $(\mu - \mu^*)^T (\mathbf{x}^* - \mathbf{x}) \leq 0$, which yields $q_{2\mathcal{I}} \lesssim -\|\mathbf{x} - \mathbf{x}^*\|^2 - (1/2)\mathbf{x}^T L\mathbf{x}$. Therefore,

$$\begin{aligned} \dot{V}^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) &\lesssim \sup_{\mathbf{h}_{\mathcal{I}}} (-\|\mathbf{x} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}]\|^2) - \frac{1}{2}\mathbf{x}^T L\mathbf{x} \\ &= -\inf_{\mathbf{h}_{\mathcal{I}}} (\|\mathbf{x} - \mathcal{P}_X[\mathbf{x} - \mathbf{h}_{\mathcal{I}}]\|^2) - \frac{1}{2}\mathbf{x}^T L\mathbf{x} \lesssim 0. \end{aligned} \quad (21)$$

For the interval-valued Lyapunov function $V^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta})$ in (14), we have $\Gamma(\mathbf{x}) - \Gamma(\mathbf{x}^*) - \langle \mathbf{x} - \mathbf{x}^*, \zeta^* \rangle \geq 0$ and $\|A\mathbf{x} - \mathbf{b}\|_1 - \|A\mathbf{x}^* - \mathbf{b}\|_1 - \langle \mathbf{x} - \mathbf{x}^*, \mu^* \rangle \geq 0$. Since $\int \partial_{\mathcal{H}}^{\delta}$

$\mathcal{F}(\mathbf{x})d\mathbf{x} = \int [\delta_i(x_i) \partial_{\mathcal{H}} \mathcal{F}_i(x_i)]_{i=1}^N d\mathbf{x} \subseteq [0, 1] \sum_{i=1}^N f_i^T(x_i)$, by virtue of the convexity of f_i^T , we know that $\partial_{\mathcal{H}}^{\delta} \mathcal{F}(\mathbf{x}) - \partial_{\mathcal{H}}^{\delta} \mathcal{F}(\mathbf{x}^*) - \langle \mathbf{x} - \mathbf{x}^*, \gamma_{\mathcal{I}}^* \rangle \gtrsim 0$. In addition, $(1/2)(\mathbf{x} + (1/2)\boldsymbol{\vartheta})^T L(\mathbf{x} + (1/2)\boldsymbol{\vartheta})$ is convex with respect to $\mathbf{x} + (1/2)\boldsymbol{\vartheta}$, it follows that

$$\begin{aligned} &\frac{1}{2} \left\langle \mathbf{x} + \frac{1}{2}\boldsymbol{\vartheta}, L \left(\mathbf{x} + \frac{1}{2}\boldsymbol{\vartheta} \right) \right\rangle - \frac{1}{2} \left\langle \mathbf{x}^* + \frac{1}{2}\boldsymbol{\vartheta}^*, L \left(\mathbf{x}^* + \frac{1}{2}\boldsymbol{\vartheta}^* \right) \right\rangle \\ &\geq \left\langle \mathbf{x} + \frac{1}{2}\boldsymbol{\vartheta} - \left(\mathbf{x}^* + \frac{1}{2}\boldsymbol{\vartheta}^* \right), L \left(\mathbf{x}^* + \frac{1}{2}\boldsymbol{\vartheta}^* \right) \right\rangle \\ &= \langle \mathbf{x} - \mathbf{x}^*, L\mathbf{x}^* \rangle + \left\langle \mathbf{x} - \mathbf{x}^*, \frac{1}{2}L\boldsymbol{\vartheta}^* \right\rangle \\ &\quad + \left\langle \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*), L\mathbf{x}^* \right\rangle + \left\langle \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*), \frac{1}{2}L\boldsymbol{\vartheta}^* \right\rangle \\ &= \left\langle \mathbf{x} - \mathbf{x}^*, \frac{1}{2}L\boldsymbol{\vartheta}^* \right\rangle + \left\langle \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*), \frac{1}{2}L\boldsymbol{\vartheta}^* \right\rangle \end{aligned}$$

which yields $w^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) - w^{\mathcal{I}}(\mathbf{x}^*, \boldsymbol{\vartheta}^*) - \Lambda^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}, \mathbf{x}^*, \boldsymbol{\vartheta}^*) \gtrsim 0$. Hence,

$$V^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) \gtrsim \frac{1}{2}\|\mathbf{x} - \mathbf{x}^*\|^2 + \frac{1}{2} \left\langle \frac{1}{2}(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*), \frac{1}{2}L(\boldsymbol{\vartheta} - \boldsymbol{\vartheta}^*) \right\rangle \geq 0 \quad (22)$$

which also shows the positive semidefinite property of $V^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta})$ and its radially unbounded with respect to $\mathbf{x}$ and $L\boldsymbol{\vartheta}$. It can be seen from (21) and (22) that the state $(\mathbf{x}, L\boldsymbol{\vartheta})$ of neurodynamic algorithm (9) is bounded. ■

Theorem 4: The state trajectory $(\mathbf{x}, L\boldsymbol{\vartheta})$ of neurodynamic algorithm (9) is asymptotically convergent, and the state solution $\mathbf{x}(t)$ converges to an efficient solution to the distributed interval-valued optimization problem (7).

Proof: For any initial point $(\mathbf{x}_0, \boldsymbol{\vartheta}_0)$, define a set $\mathcal{W} = \{(\mathbf{x}, L\boldsymbol{\vartheta}) : V^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) \lesssim V^{\mathcal{I}}(\mathbf{x}_0, \boldsymbol{\vartheta}_0)\}$, which is bounded because of the boundedness of $\mathbf{x}, L\boldsymbol{\vartheta}$ associated with neurodynamic system (9), and define

$$E^{\mathcal{I}}(\mathbf{x}, L\boldsymbol{\vartheta}) = \inf_{\gamma_{\mathcal{I}}, \zeta, \mu} \left(\left\| \mathbf{x} - \mathcal{P}_X \left[\mathbf{x} - \gamma_{\mathcal{I}} - \zeta - \mu \right. \right. \right. \\ \left. \left. \left. - L \left(\mathbf{x} + \frac{1}{2}\boldsymbol{\vartheta} \right) \right] \right\|^2 \right) + \frac{1}{2}\mathbf{x}^T L\mathbf{x}. \quad (23)$$

The following will prove that $E^{\mathcal{I}}(\hat{\mathbf{x}}, L\hat{\boldsymbol{\vartheta}}) \ni 0$ if and only if $\hat{\mathbf{x}}$ is an efficient solution of constrained distributed interval-valued optimization problem (7). Suppose that there exists a point $(\hat{\mathbf{x}}, L\hat{\boldsymbol{\vartheta}})$ such that $E^{\mathcal{I}}(\hat{\mathbf{x}}, L\hat{\boldsymbol{\vartheta}}) \ni 0$, then there exist corresponding $\hat{\gamma}_{\mathcal{I}} \in \partial_{\mathcal{H}}^{\delta} \mathcal{F}(\hat{\mathbf{x}})$, $\hat{\zeta} \in \partial \Gamma(\hat{\mathbf{x}})$, $\hat{\mu} \in \partial \|A\hat{\mathbf{x}} - \mathbf{b}\|_1$ such that

$$\begin{cases} 0 \in \hat{\mathbf{x}} - \mathcal{P}_X \left[\hat{\mathbf{x}} - \hat{\gamma}_{\mathcal{I}} - \hat{\zeta} - \hat{\mu} - L\hat{\mathbf{x}} - \frac{1}{2}L\hat{\boldsymbol{\vartheta}} \right], & (24a) \end{cases}$$

$$\begin{cases} \hat{\mathbf{x}}^T L\hat{\mathbf{x}} = 0, & (24b) \end{cases}$$

$$\begin{cases} g(\hat{\mathbf{x}}) \leq 0, \quad A\hat{\mathbf{x}} = \mathbf{b}, \quad \hat{\mathbf{x}} \in X. & (24c) \end{cases}$$

With the symmetry and positive semidefinite property of L , let $L = C^T C$ where $C \in \mathbb{R}^{Nn \times Nn}$, it follows by $0 = \hat{\mathbf{x}}^T L\hat{\mathbf{x}} = \hat{\mathbf{x}}^T C^T C\hat{\mathbf{x}} = (C\hat{\mathbf{x}})^T C\hat{\mathbf{x}}$ that $C\hat{\mathbf{x}} = \mathbf{0}$, which yields $L\hat{\mathbf{x}} = C^T C\hat{\mathbf{x}} = \mathbf{0}$. Then, $\hat{\mathbf{x}}$ is an efficient solution to problem (7). In turn, assume that $\hat{\mathbf{x}}'$ is an efficient solution to problem

(7), then there exist $\hat{\boldsymbol{\vartheta}}' \in \mathbb{R}^{Nn}$, $\hat{\gamma}'_T \in \partial_{\mathcal{S}}^{\delta} \mathcal{F}(\hat{\mathbf{x}}')$, $\hat{\zeta}' \in \partial \Gamma(\hat{\mathbf{x}}')$, $\hat{\mu}' \in \partial \|A\hat{\mathbf{x}}' - \mathbf{b}\|_1$ such that (11a)–(11c) hold, which shows $\hat{\mathbf{x}}' \in \mathcal{P}_X[\hat{\mathbf{x}}' - \hat{\gamma}'_T - \hat{\zeta}' - \hat{\mu}' - L\hat{\mathbf{x}}' - (1/2)L\hat{\boldsymbol{\vartheta}}']$ and $\hat{\mathbf{x}}'^T L\hat{\mathbf{x}}' = 0$. Thus, $E^{\mathcal{I}}(\hat{\mathbf{x}}', L\hat{\boldsymbol{\vartheta}}') = 0$. The equivalence proof is complete.

Based on Theorem 3, combined with the nonsmooth extension of Byrnes–Martin integral invariance principle [42, Th. 2.10], we know that the state trajectory $(\mathbf{x}, L\boldsymbol{\vartheta})$ asymptotically converges to the largest weakly invariant set $\mathcal{U}$ and $\mathcal{U} \subseteq \mathcal{W} \cap \{(\mathbf{x}, L\boldsymbol{\vartheta}) \in \mathbb{R}^{Nn} \times \mathbb{R}^{Nn} : 0 \in \dot{V}^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta})\}$, where $\mathcal{U}$ is bounded. When $\dot{V}^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) \ni 0$, one gets $E^{\mathcal{I}}(\mathbf{x}, L\boldsymbol{\vartheta}) \ni 0$ by (21) and (23), which follows that $\mathbf{x}$ is an efficient solution to the problem (7). Thus, the state solution $\mathbf{x}(t)$ asymptotically converges to the efficient solution set $\mathcal{O}$ to the problem (7). Further, since state trajectory $(\mathbf{x}(t_k), \boldsymbol{\vartheta}(t_k))$ is bounded, we know from Bolzano–Weierstrass theorem that there exist sequences $\{\mathbf{x}(t_k)\}$, $\{\boldsymbol{\vartheta}(t_k)\}$ and the corresponding limit points $\bar{\mathbf{x}}$, $\bar{\boldsymbol{\vartheta}}$ satisfying $\lim_{k \rightarrow \infty} \mathbf{x}(t_k) = \bar{\mathbf{x}}$ and $\lim_{k \rightarrow \infty} L\boldsymbol{\vartheta}(t_k) = L\bar{\boldsymbol{\vartheta}}$, in which $\{t_k\}$ is a strictly monotone increasing time sequence. Therefore, $(\bar{\mathbf{x}}, L\bar{\boldsymbol{\vartheta}})$ is an w -limit point associated with $(\mathbf{x}, L\boldsymbol{\vartheta})$, i.e., $\bar{\mathbf{x}} \in \mathcal{O}$. Consider a new interval-valued Lyapunov function as follows:

$$U^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) = w^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) - w^{\mathcal{I}}(\bar{\mathbf{x}}, \bar{\boldsymbol{\vartheta}}) - \Lambda^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}, \bar{\mathbf{x}}, \bar{\boldsymbol{\vartheta}}) + \frac{1}{2} \|\mathbf{x} - \bar{\mathbf{x}}\|^2 + \frac{1}{2} \left\langle \frac{1}{2}(\boldsymbol{\vartheta} - \bar{\boldsymbol{\vartheta}}), \frac{1}{2}L(\boldsymbol{\vartheta} - \bar{\boldsymbol{\vartheta}}) \right\rangle. \quad (25)$$

By a similar discussion about Theorem 3, we also get that $\dot{U}^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) \lesssim 0$. Since $U^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta})$ is continuous with respect to $\mathbf{x}$ and $L\boldsymbol{\vartheta}$, there exist $\sigma_1, \sigma_2 > 0$ such that

$$\begin{cases} \|\mathbf{x} - \bar{\mathbf{x}}\| \leq \sigma_1, \\ \|L\boldsymbol{\vartheta} - L\bar{\boldsymbol{\vartheta}}\| \leq \sigma_2, \end{cases} \Rightarrow U^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) < \epsilon, \forall \epsilon > 0.$$

By $\lim_{k \rightarrow \infty} \mathbf{x}(t_k) = \bar{\mathbf{x}}$, $\lim_{k \rightarrow \infty} L\boldsymbol{\vartheta}(t_k) = L\bar{\boldsymbol{\vartheta}}$ and also $\lim_{k \rightarrow \infty} t_k = \infty$, we conclude that there exists a corresponding $t_1 > 0$ such that $\|\mathbf{x}(t) - \bar{\mathbf{x}}\| \leq \sigma_1$, $\|L\boldsymbol{\vartheta}(t) - L\bar{\boldsymbol{\vartheta}}\| \leq \sigma_2$, $\forall t > t_1$, which follows that $(1/2)\|\mathbf{x}(t) - \bar{\mathbf{x}}\|^2 \lesssim U^{\mathcal{I}}(\mathbf{x}(t), \boldsymbol{\vartheta}(t)) < \epsilon$, $\forall t > t_1$. Therefore, the state solution $\mathbf{x}(t)$ of algorithm (9) converges to an efficient solution to problem (7). Meanwhile, the local state $x_i (i \in \{1, 2, \dots, N\})$ in neurodynamics (8) reaches consensus at the efficient solution due to $L\bar{\mathbf{x}} = \mathbf{0}$. ■

Remark 8: The theoretical proofs of Theorems 3 and 4 are analyzed based on the arithmetic of intervals, and the resulting boundedness and convergence are established on the real Euclidean space containing closed intervals, which is an extension of the neurodynamic optimization methodology. In addition, Since $U^{\mathcal{I}}(\mathbf{x}, \boldsymbol{\vartheta}) \lesssim (1/2)\langle (1/2)(\boldsymbol{\vartheta} - \bar{\boldsymbol{\vartheta}}), (1/2)L(\boldsymbol{\vartheta} - \bar{\boldsymbol{\vartheta}}) \rangle$, then $\lim_{t \rightarrow \infty} L\boldsymbol{\vartheta}(t) = L\bar{\boldsymbol{\vartheta}}$ can be obtained from a similar analysis. In distributed neurodynamics (8), for any $i \in \{1, 2, \dots, N\}$, the state output $\sum_{j \in \mathcal{N}_i} a_{ij}(\vartheta_i - \vartheta_j)$ represents a component of $L\boldsymbol{\vartheta}$ in neurodynamics (9) that is convergent, but may not satisfy consensus.

Remark 9: We conduct a rough discussion on the convergence rate of the algorithm. For each vertex i , we have $\inf_{\tau \in (0, t]} f_i^{\mathcal{I}}(x_i(\tau)) - f_i^{\mathcal{I}}_{\min} \lesssim (1/4)\|x_{i0} - y_i\|^2 + (1/2)f_i^{\mathcal{I}}(x_{i0}) + (1/2)p_i(x_{i0}) + (1/2)\|A_i x_{i0} - b_i\|_1 + (1/4)$

$\sum_{j \in \mathcal{N}_i} a_{ij}(x_{i0} - x_{j0})^2 + (1/4)\sum_{j \in \mathcal{N}_i} a_{ij}(\vartheta_{i0} - \vartheta_{j0})x_{i0} + \Lambda_i$, in which $\Lambda_i = \Psi_i(y_i) - \inf_{\tau \in (0, t]} 2\Psi_i(x_i(\tau))$, and $\Psi_i(x_i) = p_i(x_i) + \|A_i x_i - b_i\|_1 + (1/2)\sum_{j \in \mathcal{N}_i} a_{ij}(x_i - x_j)^2 + (1/2)\sum_{j \in \mathcal{N}_i} a_{ij}(\vartheta_i - \vartheta_j)x_i$, $y_i \in \arg \min 2f_i^{\mathcal{I}}$. It can be seen that the initial state of neurodynamic algorithm (8) does not affect its convergence property, but it does influence its convergence rate. This is in line with the approximate rule of neural networks, which always expects to achieve a relatively fast convergence rate when the error of the objective function is relatively large.

VI. ILLUSTRATIVE SIMULATIONS

A. Network Utility Maximization in VANETs

Example 1: In general vehicular Ad hoc networks (VANETs), network utility maximization is fraught with uncertainty, where the channel resources occupied by each vehicle sending data packets are shared by all vehicles in the transmission range of that vehicle [11], [43]. The channel resources allocated by each vehicle can be described by the transmission rate of data packets; meanwhile, the channel congestion can be controlled by adjusting the transmission rate of data packets, so that the channel load is kept within the maximum beaconing load (MBL).

Denote by v_i the i th vehicle, $\mathcal{V}$ the set of all vehicles, x_i the transmission rate of data packets of the i th vehicle, and y_i the resources allocated to the i th vehicle. $\mathcal{N}_i$ is used to represent all the neighboring vehicles of v_i , and the channel load is the sum of the transmission rate of the data packets of each vehicle and its neighbors, which is controlled below the current MBL $\mathcal{L}_i^{\max}$ to avoid channel congestion. During the transmission of data packets, all vehicles need to meet an overall effective balancing of resources, i.e., $\sum_{v_i \in \mathcal{V}} \omega_i y_i = \sum_{v_i \in \mathcal{V}} \mathcal{R}_i$, where $\mathcal{R}_i$ is the resource requirement of the corresponding vehicle, and ω_i is the effective parameter related to the allocated resources. To avoid excessive differences in the transmission rate of data packets between vehicles, the transmission rate of data packets of each vehicle should be limited to a reasonable range $[x_i^{\min}, x_i^{\max}]$. Considering the complexity and uncertainty in the intelligent transportation and transmission scenario, the utility function of each vehicle can be expressed in the form of a resource and transmission rate-dependent interval value. The goal of global optimization is to obtain the optimal transmission rate of data packets and allocated resources for each vehicle and make them reach a consensus. Therefore, such a network utility maximization problem can be described as the following distributed interval-valued optimization problem:

$$\begin{aligned} \max \quad & \sum_{v_i \in \mathcal{V}} f_i^{\mathcal{I}}(x_i, y_i) \\ \text{s.t.} \quad & x_i + \sum_{v_j \in \mathcal{N}_i} x_j \leq \mathcal{L}_i^{\max}, \sum_{v_i \in \mathcal{V}} \omega_i y_i = \sum_{v_i \in \mathcal{V}} \mathcal{R}_i \\ & x_i^{\min} \leq x_i \leq x_i^{\max}, y_i \geq 0, \forall v_i \in \mathcal{V} \\ & (x_i, y_i) = (x_j, y_j), \forall v_i, v_j \in \mathcal{V} \end{aligned} \quad (26)$$

where the local utility function $f_i^{\mathcal{I}}(x_i, y_i)$ is a concave interval-valued one, which we convert to the form of a convex function

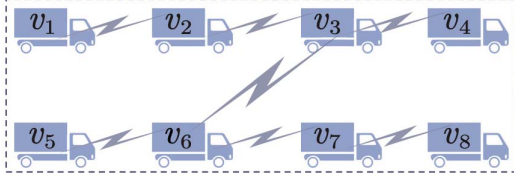


Fig. 2. Connected VANET topology in Example 1.

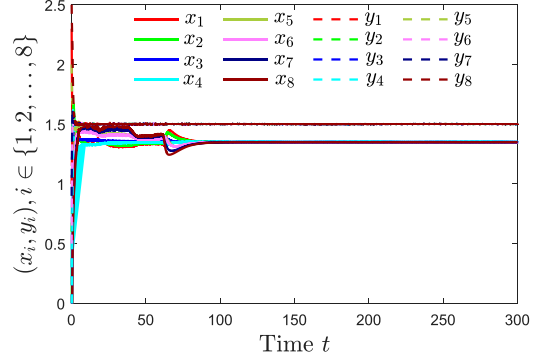
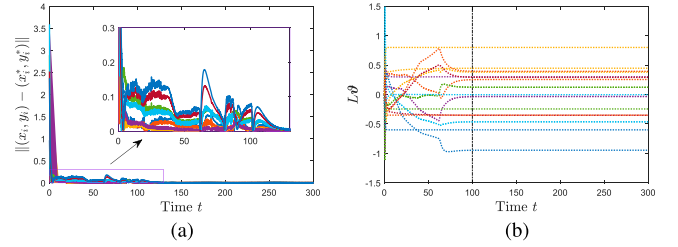
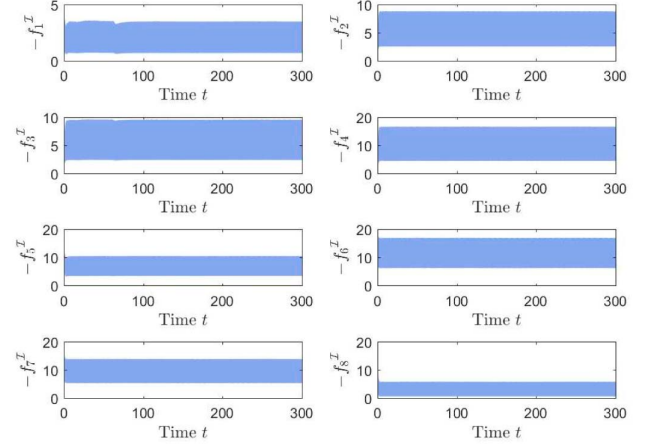
TABLE I
RELEVANT PARAMETERS IN EXAMPLE 1

Vehicle	v_1	v_2	v_3	v_4	v_5	v_6	v_7	v_8
$[a_i^L, a_i^U]$	[3,6]	[2,4]	[3,5]	[4,6]	[2.5,5]	[1.5,4]	[1,3]	[1,2.5]
$[b_i^L, b_i^U]$	[4,5]	[1,1.2]	[2,3]	[1,4]	[1,2.5]	[1,3]	[0.2,1]	[1,2]
$[c_i^L, c_i^U]$	[0.5,1]	[1,2]	[1,2.5]	[2,4]	[1.5,2]	[3,4]	[2.5,3]	[0.5,2]
$\mathcal{L}_i^{\max}$	3.6	4.8	5.5	2.7	4	6	4.5	3.2
ω_i	1	0.5	4	2	1	2.5	2	3
$\mathcal{R}_i$	2	1	4	4	2.5	5	4	1.5
$[x_i^{\min}, x_i^{\max}]$	[2,4]	[1,4]	[0,2.5]	[1,2]	[1.5,3]	[0.5,2]	[0,2]	[1,3]

so that the problem is transformed into the distributed minimization problem $(x_i, y_i) \in \arg \min(-\sum_{v_i \in \mathcal{V}} f_i^{\mathcal{I}}(x_i, y_i))$ under the constraints of data packet transmission and resource allocation.

Consider an undirected connected VANET with eight vehicles, and the topology of network is given in Fig. 2. The local utility function of vehicle v_i is described as $f_i^{\mathcal{I}}(x_i, y_i) = 1 - [a_i^L, a_i^U] \exp(-([b_i^L, b_i^U]|x_i|/4)) - (1/2)[c_i^L, c_i^U]y_i^2$, $i \in \{1, 2, \dots, 8\}$, and the corresponding parameters are shown in Table I. It is not difficult to find that $X_0 \triangleq \bigcap_{i=1}^8 X_i = \{(x_i, y_i) : x_i \in [1, 2], y_i \geq 0\}$. The communication topology between vehicles is not fully connected, and the connection weight between any two vehicles is set to 1 or 0, where 1 denotes connection and 0 denotes disconnection.

The proposed neurodynamic approach (8) is used to solve this interval-valued optimization problem. Fig. 3 shows that, driven by neurodynamics (8), the output states related to the transmission rate and allocated resources of each vehicle converge to an efficient solution where consensus is reached. Fig. 4(b) indicates an evolution curve regarding the relationship between local hidden states in neurodynamics (8), where the black dotted line indicates the consensus threshold for the convergence time of the hidden state ϑ_i . When all the components of $L\vartheta$ reach their own stable states, the corresponding local state solutions (x_i, y_i) converge to the efficient solution (x_i^*, y_i^*) of the constrained distributed interval-valued optimization problem (6). Fig. 4(a) displays the evolution of the errors between the local state solutions and the efficient solution over the execution time of the algorithm. In Fig. 5, we can see the interval range of the utility function corresponding to each vehicle, and the final interval range corresponds to the optimal transmission rate and allocated resources. Therefore, neurodynamics method (8) is feasible and effective for solving this kind of network utility maximization problem in VANETs.

Fig. 3. Evolution of consensus of output of (x_i, y_i) in Example 1.Fig. 4. Relevant evolution results in Example 1. (a) Evolution results of error functions $\|(x_i, y_i) - (x_i^*, y_i^*)\|$. (b) Output results of the components of $L\vartheta$.Fig. 5. Evolution of utility function $-f_i^{\mathcal{I}}(x_i, y_i)$ in Example 1.

B. Balancing of Power Flow

Example 2: Power flow balancing requires the active generation scheduling of each generator to meet its own load demand under the limitation of generation capacity and transmission capacity, and at the same time minimizes the generation cost of the full power network [44]. In complex transmission network, since the performance of transmission lines and generators is not always stable, it is more appropriate to describe the generation cost function with an interval-valued function.

For a power system transmission network $\mathcal{G} = (\mathcal{V}_c, \mathcal{E}_c)$, in which $\mathcal{V}_c = \{v_1, v_2, \dots, v_N\}$ denotes the set of generators, and

$\mathcal{E}_c = \{e_1, e_2, \dots, e_m\}$ denotes the set of transmission lines. If there is a transmission line $e_p \in \mathcal{E}_c$ connecting two generators v_i and v_j , then the two generators can exchange power. A reference direction of power transmission is randomly assigned to the transmission line e_p to characterize the generator-transmission line incidence matrix $W = [w_{ip}]_{N \times m} \in \mathbb{R}^{N \times m}$. If the reference direction of transmission line e_p goes to generator v_i then $w_{ip} = 1$, and $w_{ip} = -1$ if the reference direction of transmission line e_p is away from generator v_i , otherwise $w_{ip} = 0$. Denote by $\mathcal{E}_i = \{e_p \in \mathcal{E}_c : w_{ip} \neq 0\}$ the set of adjacent transmission lines of generator v_i , and $\mathcal{V}_p = \{v_i \in \mathcal{V}_c : w_{ip} \neq 0\}$ the set of adjacent generators of transmission line e_p . Such an optimization problem model for interval-valued power flow balancing can be expressed as follows:

$$\begin{aligned} \min \quad & h^{\mathcal{I}}(c, u) = \sum_{v_i \in \mathcal{V}_c} h_i^{\mathcal{I}}(c_i) + \frac{1}{2} \sum_{\substack{v_i, v_j \in \mathcal{V}_p \\ e_p \in \mathcal{E}_c}} [u_i^p, u_j^p] L_p [u_i^p, u_j^p]^T \\ \text{s.t.} \quad & c_i - \sum_{e_p \in \mathcal{E}_i} w_{ip} u_i^p - c_i^d = 0, \quad \forall v_i \in \mathcal{V}_c \\ & c_i^o \leq c_i \leq c_i', \quad u_i^o \leq u_i^p \leq u_i', \quad \forall v_i \in \mathcal{V}_c; e_p \in \mathcal{E}_i \\ & L_p [u_i^p, u_j^p]^T = \mathbf{0}, \quad \forall e_p \in \mathcal{E}_c; v_i, v_j \in \mathcal{V}_p \end{aligned} \quad (27)$$

in which $c = [c_i]_{i=1}^N \in \mathbb{R}^N$, c_i represents the generation allocation on generator v_i , and $h_i^{\mathcal{I}}(c_i)$ is the relevant generation cost function. Since there may be more than one adjacent transmission line for generator v_i , we use $\{u_i^p : e_p \in \mathcal{E}_i\}$ to denote the set including estimations of the optimal power flow on the adjacent transmission line e_p to generator v_i . Let $\mathbf{ex}\{u_i^p\}$ be the expansion form about elements in $\{u_i^p\}$, since each transmission line is traversed twice, then $u = [\mathbf{ex}\{u_1^p\}, \dots, \mathbf{ex}\{u_N^p\}]^T \in \mathbb{R}^{2m}$. By the same token, for any fixed e_p , $|\{u_i^p : v_i \in \mathcal{V}_p\}| = 2$ because any transmission line must be connected to two generators. The two inequalities represent the generation capacity constraint of generators and the transmission capacity constraint of transmission lines, respectively. The first equality guarantees the power balance of each generator, where c_i^d is the load on generator v_i .

Denote $L_p = \begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$, and the last equality guarantees that a pair of generators connected to the same transmission line must reach consensus on the power flow of that transmission line.

A distributed balancing problem of interval-valued power flow with seven-generator and six-line power grid is used as illustration. In Fig. 6, the green arrow indicates the reference direction of the corresponding transmission line. The generation cost function associated with the generator can be expressed as

$$h_i^{\mathcal{I}}(c_i) = [a_i^L, a_i^U] c_i^2 + [b_i^L, b_i^U] |c_i - \varsigma_i| + [d_i^L, d_i^U] \quad (28)$$

where the corresponding parameters are available in Table II. Also, the generation capacity limits and load demands of all generators, as well as the transmission capacity constraints of all transmission lines, are shown in Table II.

We use neurodynamic method (8) to solve this interval-valued power flow balancing problem. Fig. 7(a) shows that this method can find an optimal generation allocation for each generator after a short search. To highlight the performance

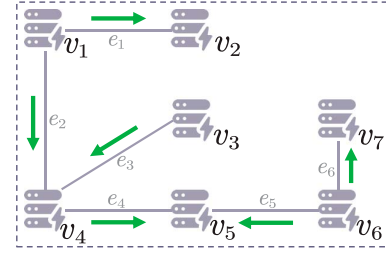


Fig. 6. Power system transmission network in Example 2.

TABLE II
RELEVANT PARAMETERS IN EXAMPLE 2

Generator	v_1	v_2	v_3	v_4	v_5	v_6	v_7
$[a_i^L, a_i^U]$	[0.5,1]	[1.5,2]	[1,2.5]	[3,4]	[0.4,1]	[1.5,3]	[1,4]
$[b_i^L, b_i^U]$	[1,2]	[0.2,1.5]	[0.6,2]	[1,2.4]	[0.6,1.8]	[2,2.5]	[1.2,2]
ς_i	3	1.5	2.5	4	1	2	4
$[d_i^L, d_i^U]$	[2.5,3]	[1,2]	[3,4.5]	[2,3]	[4,5]	[1,3]	[2,4]
c_i'	2	4	3	1	1	2	1
c_i^o	0	0	0	0	0	0	0
c_i^d	2	6	10	4	8	2	3
Transmission line	e_1	e_2	e_3	e_4	e_5	e_6	
u_i'	2	1	3	4	2	3	
u_i^o	-2	-1	-3	-4	-2	-3	

of the algorithm, we select the projected primal-dual algorithm [10] and ADMM [45] for comparison. The evolutionary results driven by these algorithms are presented in Fig. 7(b) and (c). It can be seen that the designed neurodynamic model (8) has a relative advantage. Fig. 7(d) shows the evolution of these distributed algorithms on the error terms $(1/2)\|(c, u) - (c^*, u^*)\|^2$. Meanwhile, Fig. 8 indicates that an optimal power flow for each transmission line can be found along the state solution of the algorithm, which also indicates that any pair of generators connected by the same transmission line always reach consensus on the optimal power flow of that transmission line. As can be seen from Fig. 9, an evolution of the interval-valued costs of the generators over execution time, and the final stable cost range corresponds to the optimal generation allocation. Therefore, the proposed projection neurodynamics (8) can always find an optimal generation strategy under the capacity constraints and load demands in a distributed way.

C. Pipeline Transportation of Natural Gas

Example 3: The frequent occurrence of natural disasters has caused huge losses to human life and property safety. How to distribute emergency supplies comprehensively and effectively to the disaster sites is of positive significance to the postdisaster reconstruction [46]. Suppose that in a remote region, there are s villages $\mathbf{V}_1, \mathbf{V}_2, \dots, \mathbf{V}_s$ that need the supply of natural gas, and the demand of each village is denoted as $\mathcal{D}_j (j \in \{1, 2, \dots, s\})$. In the vicinity of the region, there are r natural gas supply stations $\mathbf{S}_1, \mathbf{S}_2, \dots, \mathbf{S}_r$ that can be transported by pipeline to the s villages. Denote the amount of natural gas transmitted via pipeline from the $\mathbf{S}_i$ to $\mathbf{V}_j$ as x_{ij} , and the transmission

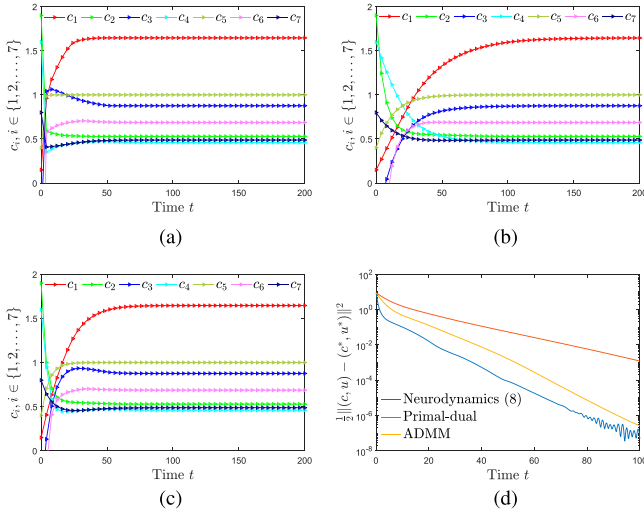


Fig. 7. Evolution of generation allocation under the influence of different distributed algorithms in Example 2. (a) Neurodynamics (8). (b) Primal-dual algorithm [10]. (c) ADMM algorithm. (d) Comparison on the error term $(1/2)\|(c, u) - (c^*, u^*)\|^2$.

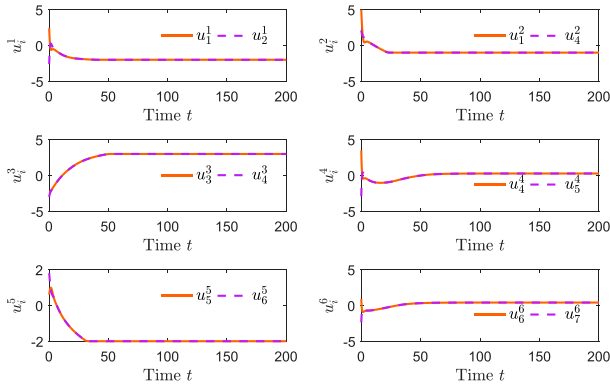


Fig. 8. Evolution results of power flow estimation on transmission lines in Example 2.

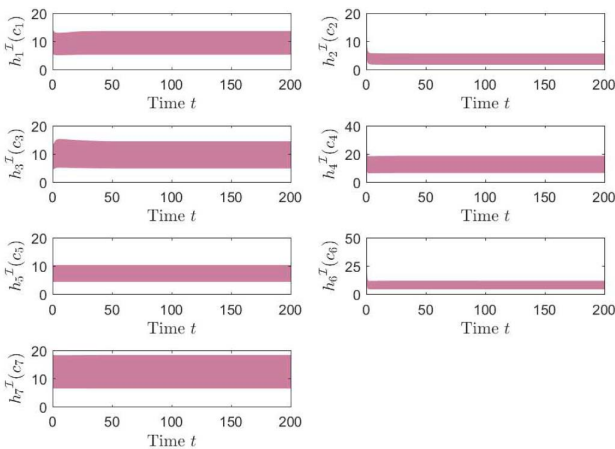


Fig. 9. Evolution of generation cost function for generators in Example 2.

cost per unit amount as c_{ij} (\$/mmbtu). Since the diameter of different interval sections of the pipeline may be different, and

the flow velocity and pressure in different interval sections are uncertain, it is more appropriate to describe the loss cost of the transmission cost per unit amount c_{ij} with the help of interval number for actual transportation, i.e., $c_{ij} \in \mathcal{I}_{\mathbb{R}}$. The balance of supply and demand should be established in the entire natural gas transportation network; that is, the sum of the amount of natural gas delivered by all supply stations to their connected villages is equal to the demand of all villages. This equilibrium condition can be mathematically formulated as

$$\sum_{j=1}^s \sum_{i=1}^r x_{ij} = \sum_{j=1}^s \mathcal{D}_j. \quad (29)$$

If the current state of supply and demand for village $\mathbf{V}_j$ is not balanced, village $\mathbf{V}_j$ will deliver its insufficient or excess demand to other neighboring natural gas supply station, and the corresponding delivery efficiency is related to the difference between supply and demand of $\mathbf{V}_j$ itself. For the pipeline transportation work of natural gas, it is usually necessary to pursue both the minimization of the total transportation cost and the maximization of the delivery efficiency. The corresponding mathematical description is as follows:

$$\begin{aligned} \min \sum_{i=1}^r \text{TC}_i^{\mathcal{I}}(x) &= \sum_{i=1}^r \sum_{j=1}^s c_{ij} x_{ij} \\ \max \sum_{j=1}^s \text{DE}_j(x) &= \sum_{j=1}^s \alpha_j \left| \sum_{i=1}^r x_{ij} - \mathcal{D}_j \right| \end{aligned} \quad (30)$$

in which α_j denotes the quantization coefficient of delivery efficiency. We combine the maximization and minimization objectives into the minimization of an interval-valued function, resulting in

$$\min f^{\mathcal{I}}(x) = \sum_{i=1}^r \text{TC}_i^{\mathcal{I}}(x) - \sum_{j=1}^s \text{DE}_j(x). \quad (31)$$

Considering the limited and safe transportation of natural gas pipelines, each natural gas supply station should carry a transportation load limit. In addition, natural gas supply may be slow at times because the villages' demand for natural gas may be an emergency. We construct the following inequality to describe that the transportation amount of natural gas is limited by the transportation load:

$$a_i \left(\sum_{j=1}^s x_{ij} \right)^2 + b_i \sum_{j=1}^s x_{ij} + d_i \leq \mathcal{R}_i \quad (32)$$

where $\mathcal{R}_i$ is the transportation load limit for the i th natural gas supply station, a_i and b_i are relevant load parameters, and d_i is the basic storage of supply station $\mathbf{S}_i$. In addition, a constraint $x_{ij} \in [\bar{x}_{ij}, \underline{x}_{ij}]$ is imposed for x_{ij} , considering the no-waste and supply pressure of each pipeline transmission line. Therefore, natural gas pipeline transportation can be modeled as the following distributed interval-valued convex optimization problem:

$$\min f^{\mathcal{I}}(x) = \sum_{j=1}^s \sum_{i=1}^r c_{ij} x_{ij} - \sum_{j=1}^s \alpha_j \left| \sum_{i=1}^r x_{ij} - \mathcal{D}_j \right|$$

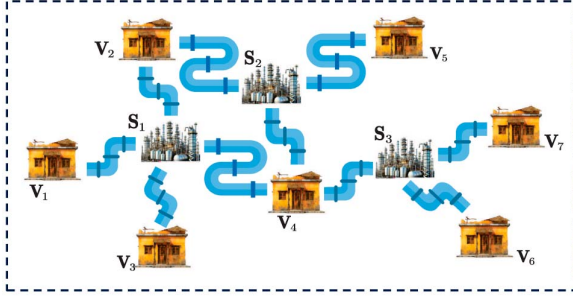


Fig. 10. Pipeline transportation lines in Example 3.

TABLE III
RELEVANT PARAMETERS IN EXAMPLE 3

Sign	Value
c_{ij}	$c_{11} = [4, 5], c_{12} = [3, 6], c_{13} = [2, 4],$ $c_{14} = [2, 2.5], c_{22} = [3, 4], c_{24} = [2.5, 5],$ $c_{25} = [5, 6], c_{34} = [3.5, 5], c_{36} = [1.5, 3],$ $c_{37} = [2, 5].$
a_i	$a_1 = 1.2, a_2 = 1, a_3 = 0.8.$
b_i	$b_1 = 1, b_2 = 0.5, b_3 = 0.4.$
d_i	$d_1 = 10, d_2 = 12, d_3 = 15.$
α_j	$\alpha_1 = 3, \alpha_2 = 2.5, \alpha_3 = 4, \alpha_4 = 3, \alpha_5 = 1.5,$ $\alpha_6 = 3.6, \alpha_7 = 2.1.$
$\mathcal{D}_j$	$\mathcal{D}_1 = 5, \mathcal{D}_2 = 4.5, \mathcal{D}_3 = 5.5, \mathcal{D}_4 = 4, \mathcal{D}_5 = 3.5,$ $\mathcal{D}_6 = 4.5, \mathcal{D}_7 = 4.$
$\mathcal{R}_i$	$\mathcal{R}_1 = 169, \mathcal{R}_2 = 114, \mathcal{R}_3 = 109.$
x_{ij}	$x_{11} \in [2.5, 4], x_{12} \in [3, 4], x_{13} \in [2.5, 4],$ $x_{14} \in [3, 4], x_{22} \in [2, 4], x_{24} \in [2, 3],$ $x_{25} \in [3, 4], x_{34} \in [2, 4], x_{36} \in [2.5, 4],$ $x_{37} \in [2.5, 4].$

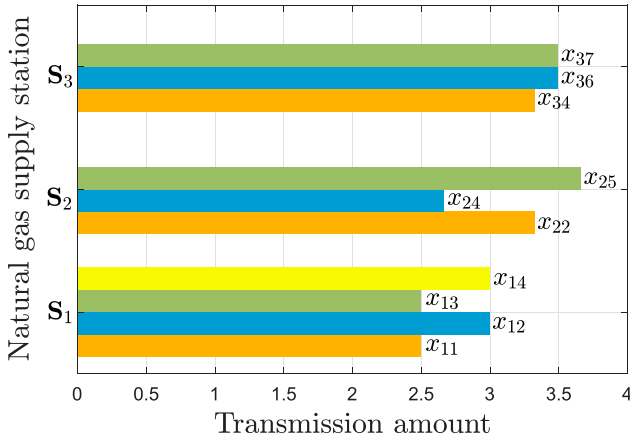
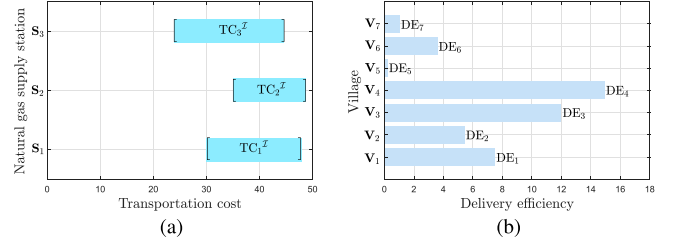


Fig. 11. Resulting transmission scheme driven by the neurodynamics (8) in Example 3.

$$\begin{aligned}
 \text{s.t. } & a_i \left(\sum_{j=1}^s x_{ij} \right)^2 + b_i \sum_{j=1}^s x_{ij} + d_i \leq \mathcal{R}_i \\
 & \sum_{j=1}^s \sum_{i=1}^r x_{ij} = \sum_{j=1}^s \mathcal{D}_j, x_{ij} \in [\bar{x}_{ij}, \underline{x}_{ij}].
 \end{aligned} \quad (33)$$

Based on the above description, we can try to use projection neurodynamics (8) to solve it.

Fig. 12. Cost and efficiency results pursued in Example 3. (a) Transportation cost of natural gas supply station S_j . (b) Delivery efficiency of village V_j .

Consider a gas pipeline transportation network with seven villages and three supply stations, as shown in Fig. 10, and the relevant parameters are given in Table III. As can be seen in Fig. 11, supply station S_1 has the largest natural gas supply to the surrounding villages, covering the villages V_1, V_2, V_3, V_4 . Supply station S_3 achieves almost a balanced supply to its connected villages, and village V_5 has an urgent need for natural gas. Fig. 12 presents the transportation cost of each supply station and the efficiency of each village, where the transportation cost is an interval value, which is caused by the uncertainty of the transportation cost and pressure on each interval section of the pipeline.

VII. CONCLUSION

In this article, we first introduced an efficient solution to describe the optimality of the solution of the interval-valued optimization problem and proposed a projection neurodynamic algorithm to solve a class of distributed convex optimization ones, where the local objective function is not necessarily gH -differentiable, and the local inequality constraint function is continuous differentiable. Under the gH -subgradient and differential inclusion system, we have achieved the construction of the neurodynamic model and discussed the efficiency condition for the solution of interval-valued optimization. The convergence of the algorithm has been proved strictly, and the obtained theory is established on the real vector space with all closed intervals, which extends the application range of the neurodynamic optimization method to a certain extent. Several application problems over the connected networks have verified the effectiveness of the neurodynamic approach. In the future, complex economic dispatch against cyber-attacks is a research work that deserves special attention [47]. We will consider using interval-valued optimization to describe such problems and design corresponding optimization methods.

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